PERCEPTION OF 3D DATA VISUALIZATIONS

by

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PERCEPTION OF 3D DATA VISUALIZATIONS

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Data visualizations are a powerful tool for the communication of complex data structures.

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A section to thank those who have supported my research and academic journey!

Table of contents

List of Figures

List of Tables

Chapter 1

Literature Review

1.1 Motivation and Background

Data visualizations play an essential role in understanding patterns in the structure of data. They allow for a programmatic mapping of data into a "picture" that can be used to gather insight from the viewer (Tufte, 2001; Tukey, 1965; Wilkinson & Wills, 2005). Since the early 20th century, researchers have been exploring the question: what makes a good graph? (Cleveland & McGill, 1984a; Croxton & Stein, 1932a; Croxton & Stryker, 1927a; Vanderplas et al., 2020a). Many attempts to answer this question have provided good recommendations, but are largely limited to the projection of graphs onto 2-dimensional (2D) surfaces. While this addresses many of the typical use cases for data visualizations, the process of creating modern statistical graphics in our 3-dimensional (3D) world is a relatively new phenomenon that does not have widespread usage (Huron et al., 2023).

Many charts exist within the confines of a 2D projected space due to the practicality of computer renderings. These computer-generated graphics are quick and cheap to produce (Tukey, 1965), which possibly contributes to their widespread usage. Before computers, hand-drawn and complex printing techniques were common production methods and took time to produce (Friendly & Wainer, 2021;

VanderPlas et al., 2019). In contrast, all data physicalizations require additional material resources, increasing the base cost to produce. Additionally, there are few modern resources that have direct pipelines from data to physical objects. One such example is the `rayshader` package (Morgan-Wall, 2024), although modifications to the resulting stereolithography (STL) file or OBJ file are needed before physicalization can take place. The cost and lack of tools for producing physical charts likely results in the majority of modern graphics being produced with 2D representations on digital screens.

Nearly every type of chart can be created by using a programmatic mapping of data and variables to aesthetics and geometries, a process sometimes referred to as the "Grammar of Graphics" (Wilkinson & Wills, 2005). This has found widespread implementation into numerous software (*SAS 9.4 ODS Graphics*, n.d.; Satyanarayan et al., 2017; Wickham, 2010). For example, consider the chart created in Figure 1.1. In this figure, the Palmer Penguins dataset (Horst et al., 2020) is supplied to the `ggplot` function from the `ggplot2` package (Wickham, 2016). The x and y axes are mapped to bill length and bill depth, with color and shape being mapped to island. The geometry is given by `geom_point`, which results in a scatterplot that shows each complete data observation. Although the chart would benefit from further customization, it was quick to produce and has instant visual insights. Other software packages can create the same chart using a similar process by specifying the axes, colors, and shapes using a point geometry.

```r
library(tidyverse)

library(palmerpenguins)

ggplot(penguins, mapping = aes(x = bill_length_mm,
                               y = bill_depth_mm,
                               color = island,
                               shape = island)) +
  geom_point()
```

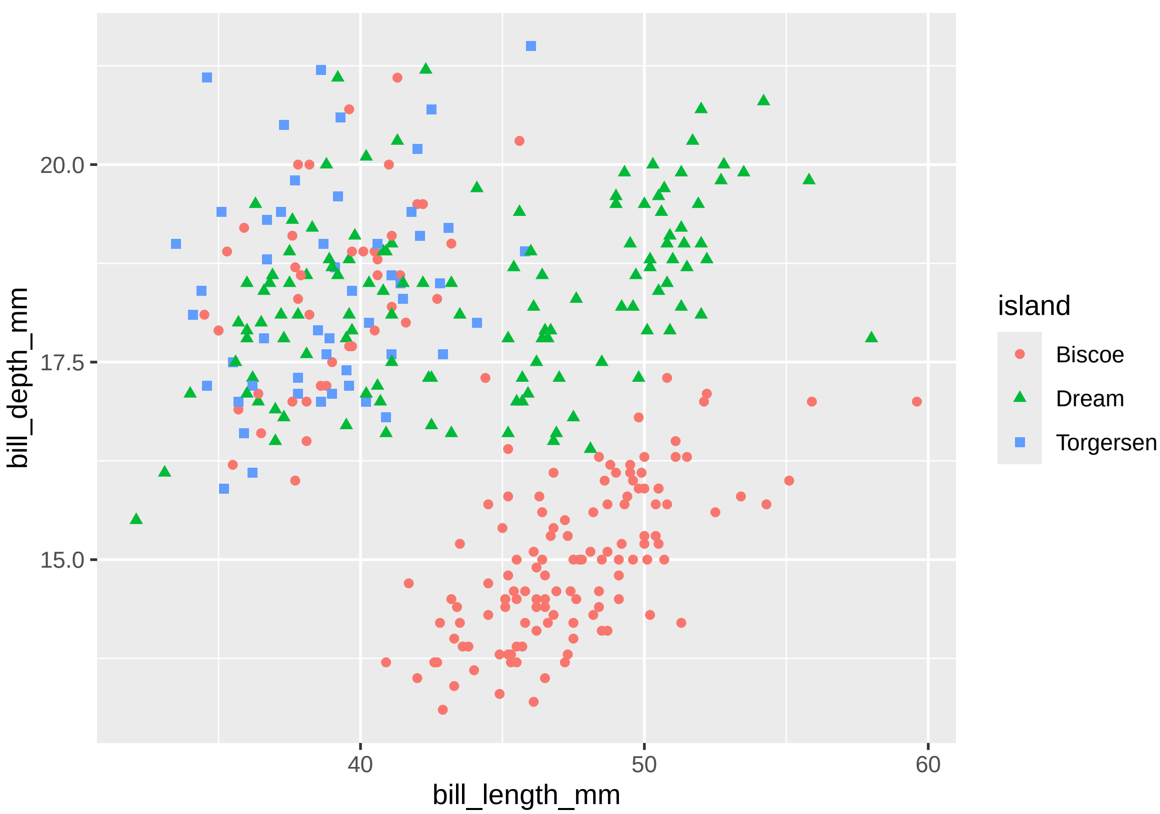


Figure 1.1: Penguins dataset (Horst et al., 2020) showing the construction of a plot using `ggplot2` in R.

While there have been many attempts to establish good data visualization practices (Tufte, 2001; Vanderplas et al., 2020a), we do not know how well these recommendations translate from 2D projections into our 3D world. The advent of

Figure 1.2: 3D-printed treemap created displaying Gross National Income for 130 countries (Huron et al., 2023, pg. 217). The two 3D prints on the bottom have additional information overlaid on top of the treemap as 2D displays.

3D printing allows us to precisely construct the 3D charts that have been widely studied through their 2D representations (Barfield & Robless, 1989a; Croxton & Stein, 1932a; Zacks et al., 1998). With limited research on 3D-printed data manifestations, we explore some of the first empirical findings of physical 3D statistical graphics.

1.2 Overview of Physical 3D Visualizations

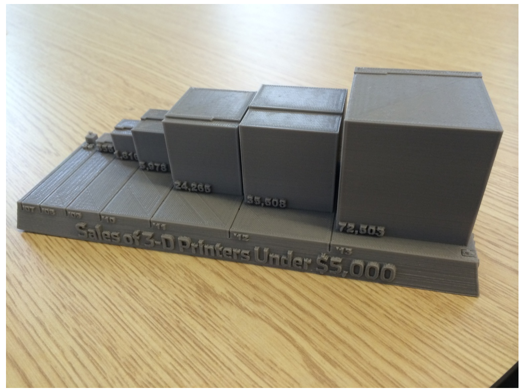
Data can be expressed in the physical world through a wide variety of means. In many cases, this can take on artistic expressions, such as encoding information into crocheted blankets or wooden sculptures (Huron et al., 2023). While certainly providing information about the data, these informational art pieces are largely left as display items to promote conversation into the underlying dataset.

Perhaps one of the most affordable types of data physicalization is through the use of 3D printing. Aside from the startup cost of a 3D-printer, material and labor costs are relatively low. Rolls of PLA filament can be acquired for around $20 and can be used to create multiple charts. Print times vary depending on the size and complexity of the graphic, but smaller charts can be produced in approximately 2 hours. While needing much more time to produce than computer-generated visualizations, the amount of hands-on labor during the 3D-printing process is mainly limited to the time it takes to create the print file.

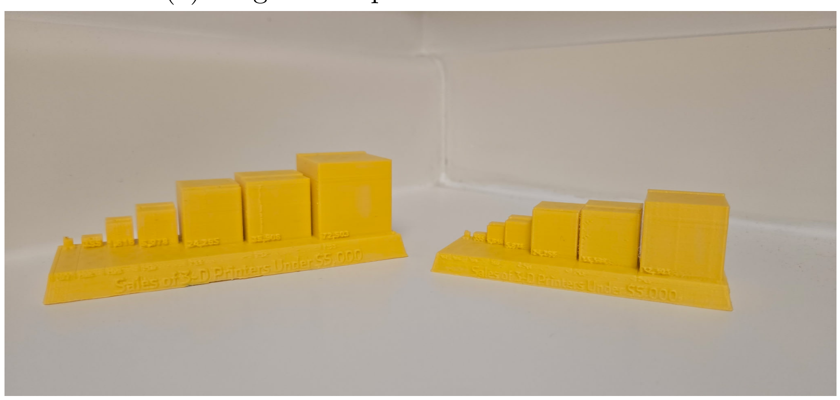
Using 3D printers to create statistical graphics has typically been reserved for niche purposes to promote engagement of the displayed data. One of the earliest 3D-printed charts was shared in 2014 by the Wall Street Journal, where the print was used to discuss the rising popularity of 3D printers (Thingiverse.com, n.d.; **graphicsHowDoes3D?**). It is rare to find these types of charts in practice, where images of the charts are often posted to social media or blog posts. This results in limited access to the physical charts, even though source print files can be shared through sites such as `thingiverse.com` and `printables.com`, as well as through GitHub.

1.2.1 Creating 3D-Printed Visualizations

The popularity of the 3D printer exploded in the early 2010s as the technology became increasingly available and cheaper. This has led to increased use of 3D printing in many focus areas, including healthcare (Dodziuk, 2016) and engineering (V et al., 2023), but has seen only limited use for statistical graphics. The creation of 3D-printed statistical graphics involves generating an STL file and using it within 3D printing software to produce the final print file. This file can then be transferred to a 3D printer, where the chart can be materialized. Many software programs have the ability to create 3D statistical graphics, but lack the ability to easily export the



(a) Original 3D printed chart from WSJ



(b) Replicated print created in 2025

Figure 1.3: Figure 1.3a shows the original WSJ print from 2014. Figure 1.3b shows two prints created with a Flash Forge Finder 3D Printer, where the smaller chart took 2 hours to print and the larger chart took 6 hours to print.

graphics into files suitable for 3D printing.

Numerous software programs can create digital renderings of 3D charts. Excel (Microsoft Corporation, 2018) has native support for creating charts with 3D depth cues, but requires add-ins to produce charts using 3 axes. R (R Core Team, 2025) has several options for creating 3D charts that have data on 3 axes (Morgan-Wall, 2024; e.g., Murdoch & Adler, 2023; Sievert, 2020), but lacks support for creating depth charts. SAS 9.4 (*SAS 9.4 ODS Graphics*, n.d.) has options such as PROC GCHART for adding depth cues and PROC G3D for surface charts. Other popular software programs contain similar capabilities, such as JavaScript libraries, MATLAB, and Python.

While a number of tools exist for creating 3D data visualizations, the pipeline of getting these charts into files compatible with 3D printing is not widely automated. For example, R has some options using the `rgl` (Murdoch & Adler, 2023) and `rayshader` (Morgan-Wall, 2024) packages, but both packages have limited tuning parameters for their output files. Another option is to use 3D software, such as OpenSCAD (Kintel, 2023), to manually create the charts, but these software programs lack the ability to directly integrate statistical information in the formulation of the charts. Without easy tools for creating 3D print files from the original data source, the process of created 3D-printed charts requires extensive manual effort to create the print file.

Another consideration for 3D-printed charts is the inclusion of text labels. These labels, and text annotations, are essential in communicating the context of the data (Homestead, 2021). With single-filament 3D printers, text can be incorporated in three primary ways: (1) embossing, in which the text is raised above the surface of the print; (2) engraving, in which the text is recessed into the surface; or (3) applying external labels, such as stickers or ink. Multi-filament 3D printers have the additional option to use a different color for text labels, which can be applied to be either flush with the surface of the chart or embossed for tangibility. However, Munzner (n.d.) cautioned that the text of 3D charts may suffer from legibility issues due to the distortion of the text when viewed at an angle. Few recommendations exist for the incorporation of labels on data physicalizations, but the use of labels should complement the construction of the chart (Sauvé et al., 2022).

1.3 Bar Charts in Research and Practice

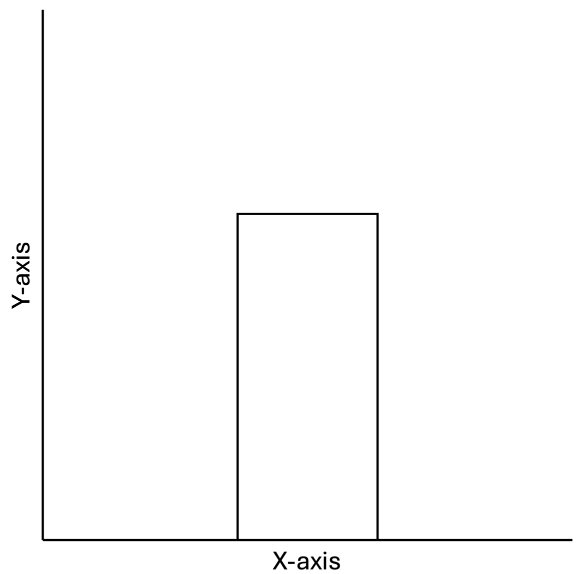
One of the most common types of charts is the bar chart, characterized by the use of rectangular geometries in a 2D space. Two variables are mapped to the geometries of the bar chart, where one axis is dedicated to a categorical variable of nominal or ordinal nature, and the other axis is for a continuous variable to denote magnitude or a response. These charts have been used over 200 years and remain a popular choice for many modern data visualizations (Friendly & Wainer, 2021). The use of 3-dimensional elements in bar charts typically involve converting the rectangular bars into rectangular prisms. Figure 1.4 shows an illustration of this transformation.

The naming convention for these types of bar charts is not consistent. Many researchers in the visualization community call these 3D bar charts due to the portrayal of three-dimensional space (Fischer, 2000; e.g., Zacks et al., 1998). Other times, the lack of information in the third dimension results in the charts referred to as 2.5D charts (e.g., Tractinsky & Meyer, 1999). To be consistent, we will refer to this style as “3D bar charts”.

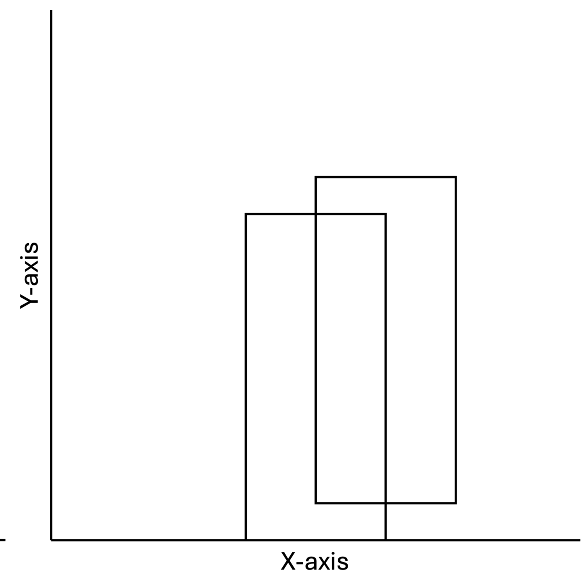
1.3.1 Against 3D Bar Charts

Tufte (2001) called the use of 3D elements a “fake perspective” in his description of “chart junk”, which is where visual elements add clutter to the data visualization. Other researchers (Stewart et al., 2009; Zacks et al., 1998) have termed the perspective effect “extraneous” when referring to depth cues. The reasoning behind this is fairly intuitive: why include additional noise when simpler alternatives exist?

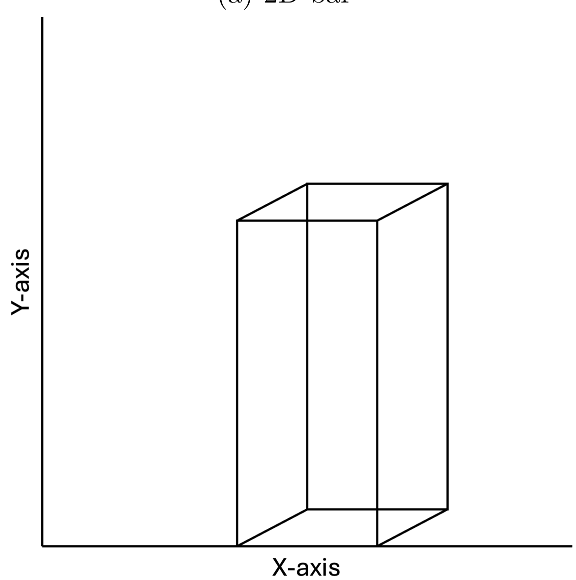
The seminal work of Cleveland & McGill (1984a) theorized that encoding



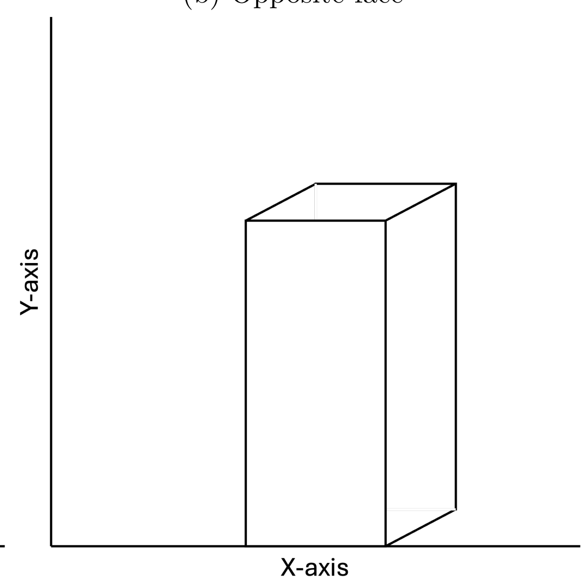
(a) 2D bar



(b) Opposite face



(c) Connecting faces



(d) Solid object

Figure 1.4: Construction of depth cues for a 2D bar chart. In some cases, bar are constructed such that the y-axis corresponds with the front face of the prism (as shown). Other 3D bar charts may have the y-axis aligned with the back face.

information in volumes would provide worse numerical estimation than for encodings using positions, lengths, and areas. Their reasoning stems from Stevens' Power Law (Stevens, 1957), a psychophysics formulation that estimates a perceived stimulus magnitude p with the actual magnitude a by $p = ka^{\alpha}$. The estimation of α provides guidance as to which types of stimuli are most subject to distortion from their true scale when $\alpha = 1$. Stevens (1957) estimated α to be approximately 1.1 for visual length and as low as 0.9 for visual area. While the 3D bar charts do not lose their 2D encodings, they do gain depth, and thus volume encodings. The effect of volume versus area comparisons was noted earlier by Croxton & Stein (1932a), where accuracy for ratio estimations was worse for cubes as compared to bars and circles. When considering the dimensionality, comparisons of lower dimensions tend to be less distorted to human perception than higher dimensions.

From an artistic point of view, depth cues can obscure some elements of the chart, resulting in a decrease in readability. For 3D bar charts, this often occurs from the volumetric elements partially covering the response axis. Consider Figure 1.5 where the y-axis corresponds to the back face of the rectangular prism. In this case, the front and right faces do not directly correspond with the axis, thus making it more difficult to extract values from the chart.

1.3.2 For 3D Bar Charts

The primary motivations for using 3D bar charts are preference and memorability. Levy et al. (1996) argued that depth cues can be useful in these contexts. In three experiments involving over 100 psychology students, participants were asked to select from a set of 2D and 3D charts for different tasks. Their results suggested that students more often preferred 3D charts when the goal was to enhance memory of the chart, while preferences were about evenly split between 2D

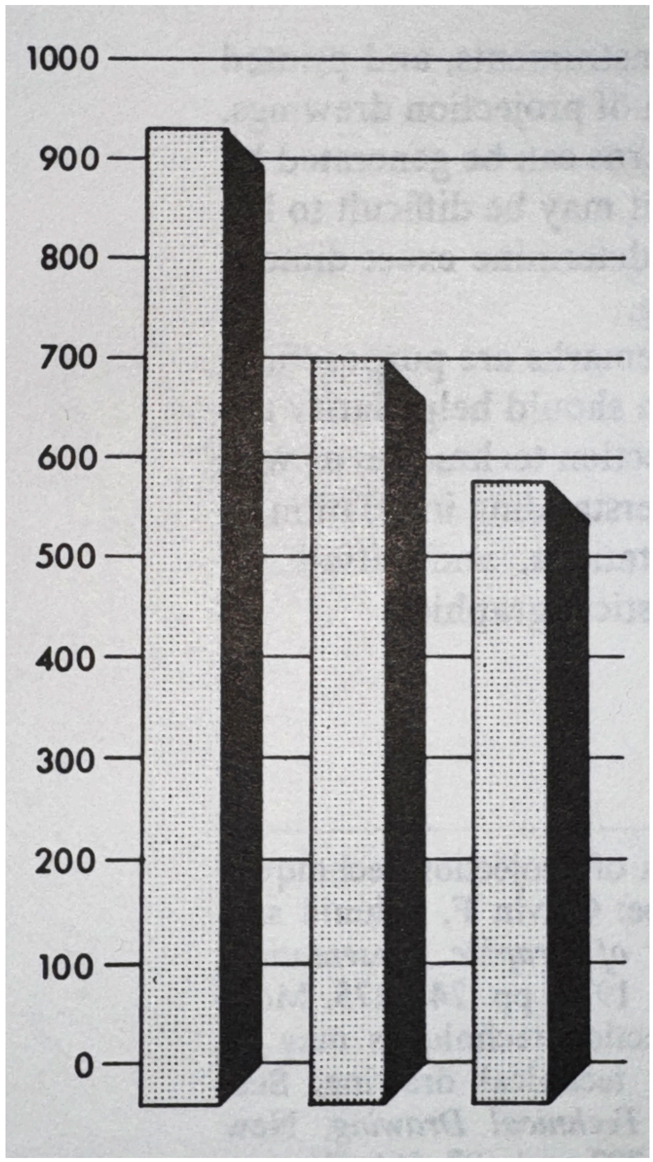


Figure 1.5: Figure from Schmid (1983) when discussing 3D visualizations. The viewing angle for this chart would likely have viewers overestimate the values of the bars by lining up the front face with the y-axis.

and 3D when the task was to present information to others. Levy et al. (1996) also suggested that the depth embellishments in 3D graphics may boost memorability by making charts more visually distinctive.

The primary motivations for using 3D bar charts are preference and memorability. Embellishments through the use of 3D elements could be interpreted as more impressive, and thus better for the final polished chart. If the chart is visually appealing, it may increase the memorability by having those distinctive elements. A study of chart tasks was conducted by Levy et al. (1996), where it was suggested that 3D charts are preferred for these reasons. Another study conducted by Fisher et al. (1997a) suggested similar preferences. In each of these studies, 2D charts were preferred for extracting information, and approximately half of participants preferred 3D charts for displaying information. Levy et al. (1996) speculated that the preference for 3D charts was to make the displays stand out to the viewer and thus more memorable.

The idea of 3D charts increasing memorability aligns with broader findings on "chart junk," where non-data elements are suggested to aid recall of information found in visualizations, though often at the cost of longer processing times (Bateman et al., 2010; Borgo et al., 2012; Peña et al., 2020). Similar increases in processing time have been reported specifically for 3D bar charts (Fischer, 2000; Siegrist, 1996; Stewart et al., 2009). The trade-off between information extraction and memorability provides some evidence that 3D bar charts may have context-specific roles in data visualizations.

The persistence of 3D bar charts reflects the influence of artistic preferences in visualization. The added depth can make results appear more striking, giving the impression of visual weight. Such embellishments often appeal to audiences by enhancing visual impact, even when they add little to the clarity of the data itself.

1.3.3 Conflicting Empirical Studies

Although Tufte (2001) argued against the use of 3D charts whenever possible, empirical studies involving direct comparisons of 2D and 3D bar charts have shown mixed results. In general, these studies focus on metrics of accuracy and find that 3D bar charts tend to be either less accurate or just as accurate as their 2D counterparts. Another common metric is response time, where 3D charts tend to have longer response times. Overall preference for one style of chart over another is a subjective measurement, but has shown mixed results for the two styles of charts.

1.3.3.1 Accuracy

Accuracy is a common metric in studies comparing 2D and 3D bar charts, typically measured by either extracting a single numeric estimate or making comparisons between two values. Here, the style of chart with lower errors are considered to be the better chart. Many studies used controlled environments for constructing stimuli, often displaying one or two stimuli at a time, or controlling non-target stimuli. In practice, charts are often more complex, displaying many stimuli without drawing attention to particular values.

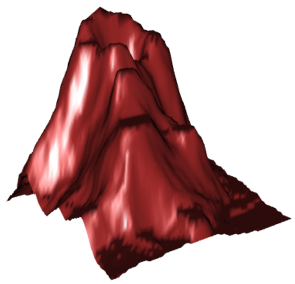
When only measuring accuracy, 3D bar charts tend to be less accurate than their 2D counterparts (Stewart et al., 2009; Zacks et al., 1998). However, this result is less prominent when introducing time delays (Zacks et al., 1998, Experiment 2) or easier task complexity (Stewart et al., 2009). Siegrist (1996) noted that bar positions and sizes also affected performance accuracy, emphasizing that other factors contribute to accuracy. It is also important to note that metrics of accuracy are not conclusive, where sometimes there is no evidence of a difference between 2D and 3D bar charts (Melody Carswell et al., 1991).

Of course, there are other less-studied factors that could affect perceptual judgments of accuracy in 3D bar charts. In addition to viewing parameters found in 2D bar chart studies (Fischer et al., 2005; Rice et al., 2024), adding depth cues are also subject to viewing angles, amount of depth, and field of view parameters. Viewing angle, with respect to the axis, can mislead the reader into overestimating or underestimating numerical quantities Figure 1.5. Zacks et al. (1998) did not find a significant effect for the amount of perception depth, but noted a trend such that increased exaggeration of depth lowered their accuracy metric. Lastly, field of view is fixed for the viewer, but can widely distort 3D renderings Figure 1.6. These factors add additional complexity, but are not widely studied for the use of statistical graphics.

1.3.3.2 Response Times

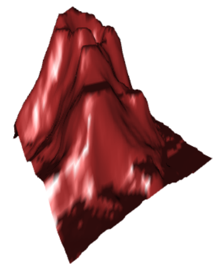
The speed in making perceptual judgments is another common metric in studies comparing types of charts. In theory, charts where questions can be answered more quickly implies that the chart is better at communicating information. However, response time is a poor metric in situations of exploratory data analysis and long term interactions with complex graphics (Vanderplas et al., 2020a). When a question of interest is known in advance, response time becomes a more viable metric.

Depth cues from 3D bar charts tend to increase the amount of time required to answer questions (Fischer, 2000; Siegrist, 1996; Stewart et al., 2009). Intuitively, additional complexity should require additional time to process. For 3D bar charts, viewers need to align the response axis with the rectangular prisms in order to determine the approximate magnitude of the bar. Consider Figure 1.5, where the y-axis denotes magnitude with respect to a hidden face of the bar. In order to



(a) FOV: 30 degrees

(b) FOV: 60 degrees



(c) FOV: 90 degrees

(d) FOV: 120 degrees

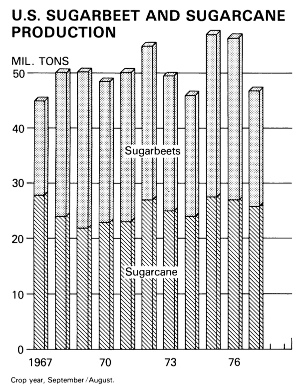
(e) FOV: 150 degrees

Figure 1.6: Multiple charts of the volcano dataset (R Core Team, 2024) rendered using the `rgl` package (Murdoch & Adler, 2023) in R. Each chart has a different field-of-view parameter, with the amount of distortion increasing as the field-of-view (FOV) increases.

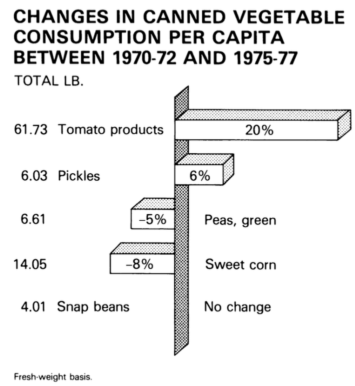
extract a numerical quantity, the viewer first has to align the axis with the correct face of the prism before processing the magnitude. The additional interaction of mentally aligning with the chosen viewing parameters results in 3D charts needing additional time to process.

1.3.3.3 Preference

The debate between whether to use 2D or 3D graphics stems from a wider adoption of 3D charts in practice. Computer graphics have made it easy to produce



(a) Bar chart with strictly positive values



(b) Bar chart with negative values

Figure 1.7: Two examples of 3D bar charts found in publication (*1978 Handbook of Agricultural Charts. Enlargements / [u.s. Department of Agriculture]*, 1978). This left image is a stacked bar chart, using shadings to indicate sugarbeats and sugarcane. The image on the right is a bar chart with positive and negative values of changes in canned vegetable consumption.

charts of either dimensionality, resulting in creative liberties for the creation of the chart for publication. Figure 1.7 showcases two examples of 3D bar charts from the 1978 Handbook of Agricultural Charts (*1978 Handbook of Agricultural Charts. Enlargements / [u.s. Department of Agriculture]*, 1978) where other options exist for displaying the data. Schmid (1983) (Chapter 8) also showcases many other 3D charts published in official reports. Evidently, the persistence of 3D bar charts has shown preferential use by maintaining a presence in many software programs, including the widely used Microsoft Excel.

Preference for data visualizations can be categorized into visual appeal and overall usefulness. Visual appeal is where perceptual tasks are not a priority, and the goal is to have an image that is pleasant to the viewer. Charts chosen for visual appeal may not accurately convey each individual observation, but rather emphasize key trends of the data or promote engagement with the visualization. Usefulness, on the other hand, is a rating of how effective the viewer thought the chart was for answering specific questions. This often comes in the form of comparing multiple

Three examples of chart preference.

values, where accuracy in the comparisons is an essential component of the chart. The purpose of the chart can be an influential factor in deciding which chart type to use for displaying a dataset (Levy et al., 1996).

When not tasked with estimating values, 3D bar charts tend to be more visually appealing (Fisher et al., 1997a; Levy et al., 1996; Stewart et al., 2009). This may be in part due to the forced illusion of depth which creates additional embellishments that would increase the impression and memorability of the chart, as seen by other types of Tufte's "chart junk" (Borgo et al., 2012; Peña et al., 2020). Despite the visual appeal, 2D charts tend to be chosen for preference and ease of use when making numerical estimations (Levy et al., 1996; Stewart et al., 2009). The contrast between the visual appeal of 3D bar charts and the utility of 2D bar charts is interesting to note since we would intuitively expect charts that are easier to read as being the optimal chart. This does not seem to be the case for the dimensionality of bar charts.

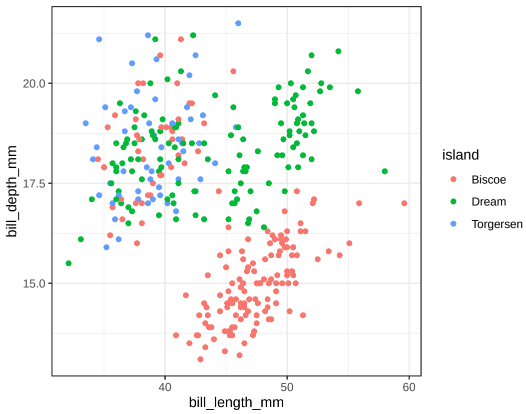
1.4 Heatmaps in Research and Practice

When creating charts for three or more variables, it becomes increasingly complex to visualize data (Grinstein & Trutschl, 2001). One option is to map the additional variables to aesthetics using gestalt principles (Vanderplas et al., 2020a), which includes making use of color, size, and/or shapes to denote information. However, not every aesthetic is compatible with every type of variable. Color can be used to map either continuous variables with gradient scales or categorical variables with palettes. Size is typically reserved for continuous variables, where larger values correspond to a larger quantity. Shapes tend to have finite distinguishable representations in order to represent categorical variables. Examples of good and poor cases for each of these aesthetics are presented in Figure 1.8.

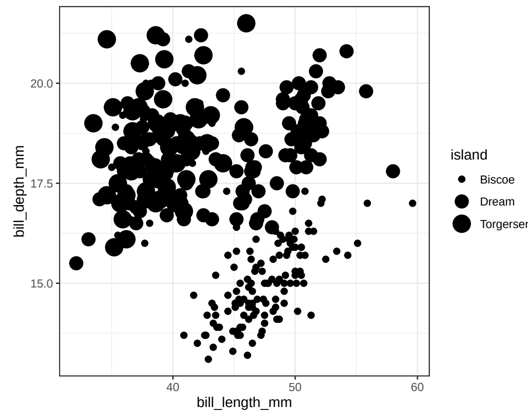
Heatmaps are a type of chart that can be used with two nominal or ordinal variables, and a continuous response variable. In two-dimensional formats, gradient fills are often used to denote the magnitude of the responses, utilizing a gradient scale. When represented in three dimensions, height is the primary mapping of the response of the heatmap. Due to the height aesthetic, 3D heatmaps are sometimes referred to as “height maps”, particularly when spatial coordinates are involved. Since the color and/or height of the response variable can only be mapped once to each combination of x and y coordinates, heatmaps are generally used to summarize responses, such as averages or single-point observations. This makes heatmaps especially useful for visualizing patterns and trends across large datasets in a compact and interpretable format.

1.4.1 Empirical Studies of 2D and 3D Heatmaps

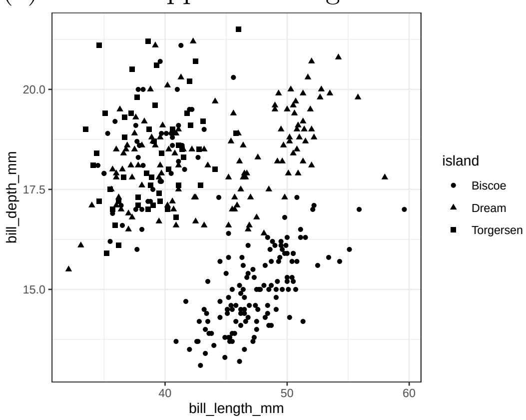
Unlike 2D and 3D bar charts, the dimensional comparison for heatmaps has not been widely studied. This may be attributed to the complex designs of heatmaps, where there are many possible variables to consider. For example, the complex nature of color scales having an influential role in the perception of 2D heatmaps (Breslow et al., 2009; Liu & Heer, 2018a; Reda & Papka, 2019). It has also been observed that neighboring values may influence the ability to extract values, either by distraction or visual illusions [zacks1998; VanderPlas & Hofmann (2015)]. The dimensionality of heatmaps requires the additional consideration of mapping color scales into heights, which is a process that has no literal translation. Depending on the scaling when mapping color to height, features may become overly smoothed or overly exaggerated. An example of this mapping discrepancy is shown in



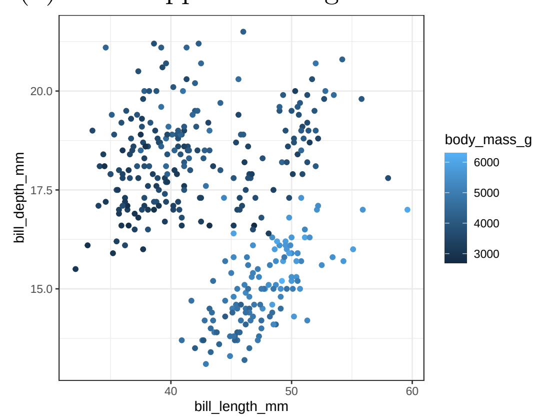
(a) Color mapped to categorical variable.



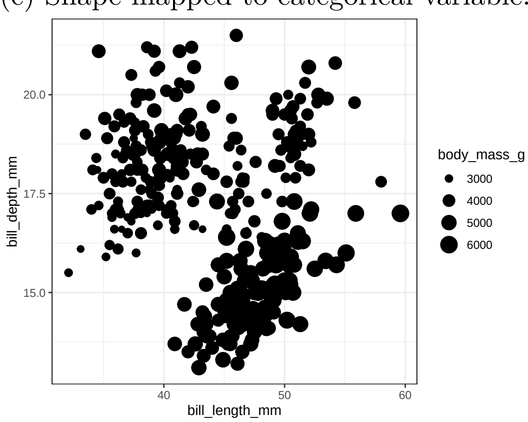
(b) Size mapped to categorical variable.



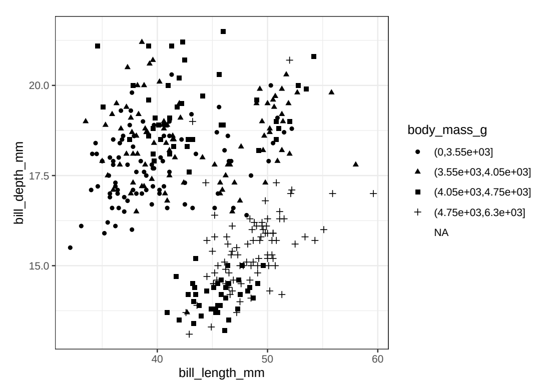
(c) Shape mapped to categorical variable.



(d) Color mapped to continuous variable.

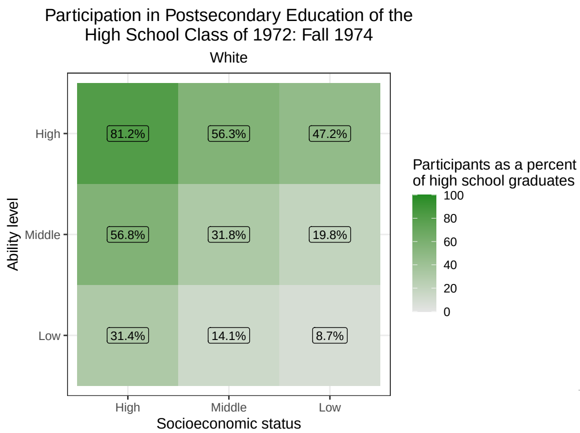


(e) Size mapped to continuous variable.

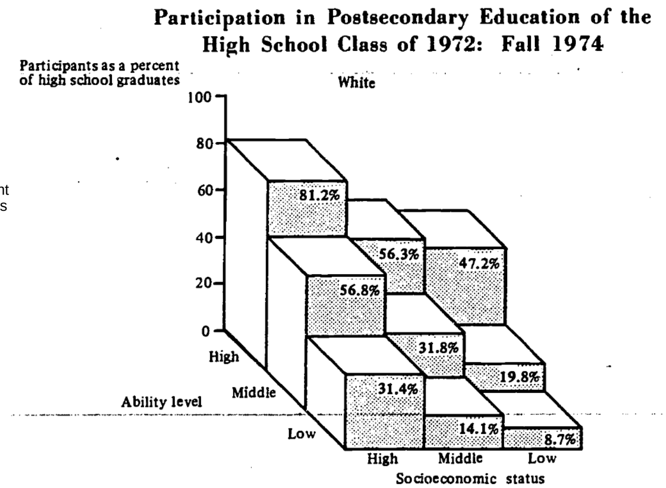


(f) Shape mapped to continuous variable.

Figure 1.8: Examples of color, size, and shape for scatterplots of the penguins dataset (Horst et al., 2020).



(a) 2D heatmap



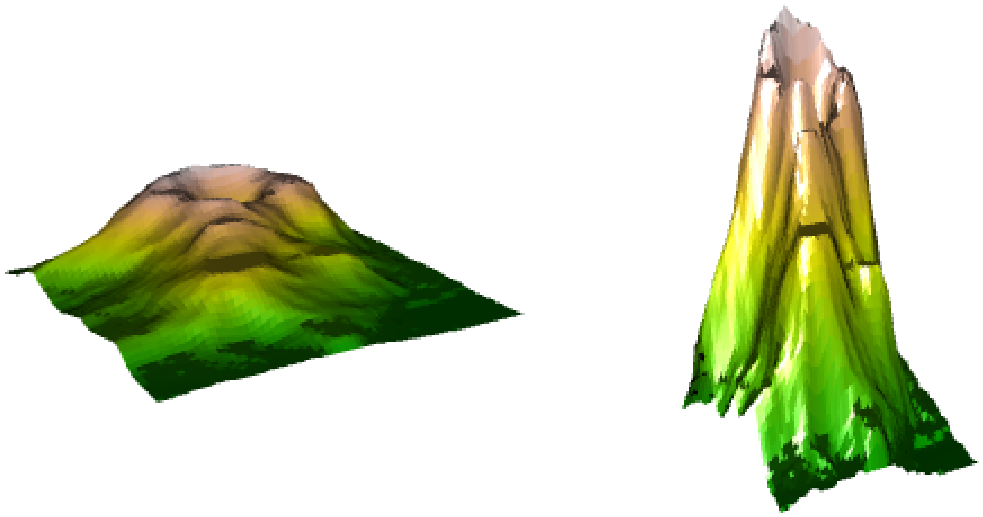
(b) 3D heatmap

Figure 1.9: Heatmaps created with data from for Educational Statistics et al. (1977). The 2D heatmap represents the percentage of high school graduates participating in postsecondary education using a gradient color scale, whereas the 3D heatmap uses height. In each chart, there is an increasing association with postsecondary education with ability level and socioeconomical status.

Figure 1.10. We expect that these numerous design choices contribute to the lack of research on 2D and 3D heatmaps.

In 2D heatmaps, the choice of color palettes can influence the perception of numerical estimations (Liu & Heer, 2018b; Molina Lopez et al., 2023; Zhang et al., 2023a). Some gradient scales are easier to extract values from than others, and other gradient scales are better at displaying contrasts between values. Additionally, the use of colors requires accessibility considerations for people with color deficiencies. The relationship between color and 2D heatmaps possibly contributes to the difficulties of direct translations of 2D heatmaps into 3D heatmaps.

Only a few studies have directly compared the dimensionality of statistical graphics (Jeong et al., 2025; Kraus et al., 2020). Conducted primarily in virtual reality, these studies suggest that 2D and 3D charts each offer advantages depending on the task. Two-dimensional charts generally produced higher interpretation



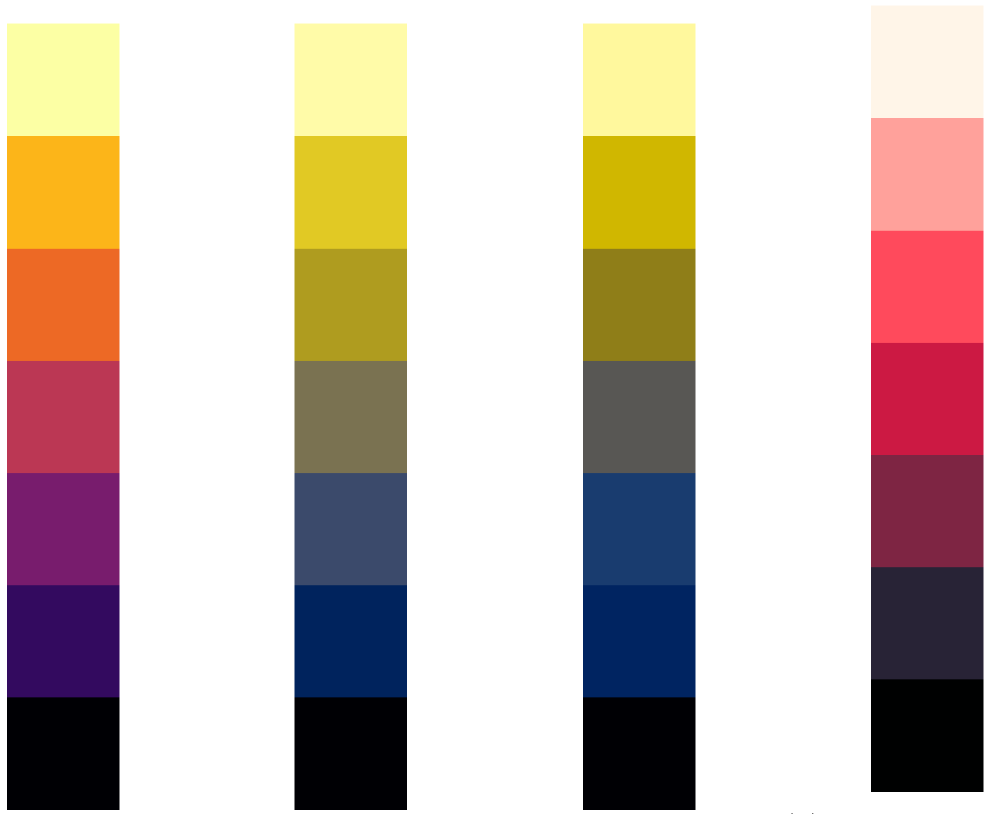
(a) Flatter height scale (b) Stretched height scale

Figure 1.10: Two figures representing the volcano dataset (R Core Team, 2024) using different height scaling factors. Both figures use the same color scale, but there is no “correct” translation of color into the measurable height.

accuracy and better recognition of overall distributional patterns. In contrast, three-dimensional charts were more effective for estimating single values and supporting long-term recall. Taken together, these findings highlight dimensionality as an important consideration when selecting visual displays for specific analytic goals.

1.5 State of Data Physicalization in 3D Statistical Graphics

While numerous studies explored the dimensionality of statistical graphics in 2D renderings, only a few studies empirically evaluated their effectiveness in 3D environments. Jansen et al. (2013a) conducted one of the first studies with physical 3D heatmaps and focused on task completion times for estimations, suggesting that physical 3D heatmaps were better for extracting information. Jansen et al. (2013a)



(a) Original palette (b) Deutan (red-green) (c) Protan (red-green) (d) Tritan (blue-yellow)

Figure 1.11: Effect of color deficiencies on the inferno palette from (**viridis?**).

also measured errors in estimations, but did not detect differences attributed to chart types. A series of experiments by Simon Stusak evaluated the memorability of digital and physical bar charts, which suggested that physical 3D bar charts may improve the ability to recall information (Stusak et al., 2015a; Stusak et al., 2016). To date, physical statistical graphics is a niche area in data visualizations and is largely unexplored.

1.6 Methods for Evaluating Visualization Effectiveness

1.6.1 What makes a good chart?

There are many aspects for designing good data visualizations. Vanderplas et al. (2020a) discussed various approaches to testing graphics, including numerical estimations, speed, and error rates. These metrics are fairly common to measure when comparing 2D and 3D charts (Cleveland & McGill, 1984a; Croxton & Stein, 1932a; Hughes, n.d.), but they do not represent the full picture of what makes one type of chart better than another. As observed by Levy et al. (1996), the objectives of the chart can determine which type of chart should be used. While many metrics exist, the practical limitation of testing generally allows for only a few metrics to be tested at a given time.

Many studies involving statistical graphics tend to rely on numerical estimations for recommending data visualization practices. This is particularly useful given the role that data visualizations play in the understanding of underlying datasets, which is to understand data structures and relationships (Tukey, 1965). Successful visualizations should be able to translate tables into figures that rapidly convey values and patterns found within the data structures. In this research, we focus primarily on numerical estimations while also recording task completion times.

1.6.2 Designing studies for numerical estimations

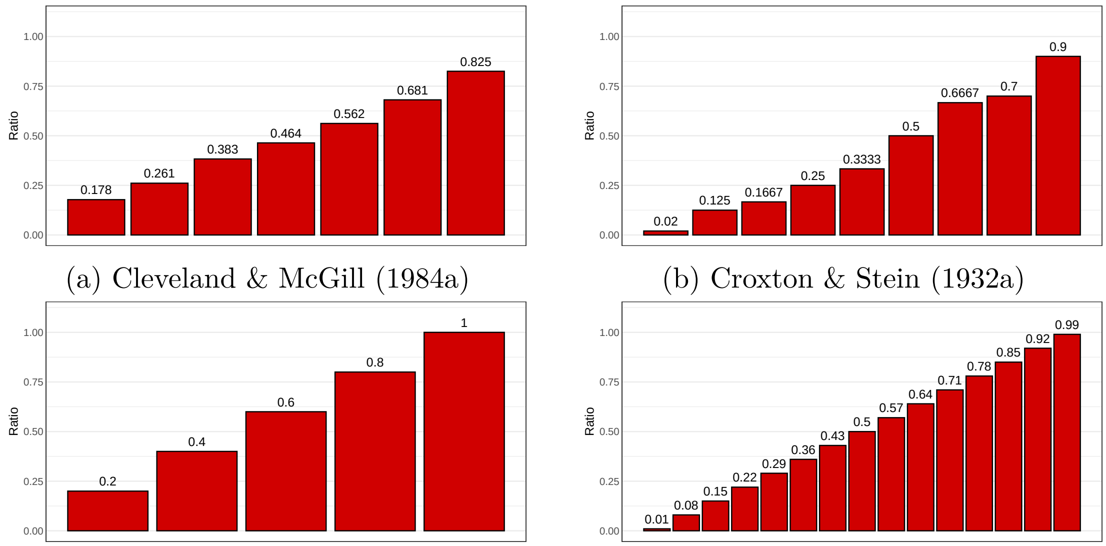
Many studies in graphical perception follow the same general format as Cleveland & McGill (1984a). Their work has been credited with over 2,500 citations, which indicates the importance of testing features of statistical graphics. Since 1984, the methods of analyzing and conducting studies on statistical graphics have evolved alongside statistical methods and technological progress (e.g., Hofmann et al., 2012; Vanderplas et al., 2020a), but the structure of experimental designs involving numerical estimations tends to remain similar. Tasks typically involve the comparison of two stimuli, where the question is “how much bigger/smaller is one value compared to the other?” The phrasing of this question needs careful consideration since mental subtractive processes are perceived differently from mental ratio processes (Hagerty & Birnbaum, 1978; Veit, 1978). Our work, like many others in the InfoViz community, follow a procedure similar to Cleveland & McGill (1984a).

Cleveland & McGill (1984a) asked participants two questions when making comparisons between two values on a chart: which of the two values was the smaller, and what percentage the smaller was of the larger. The first question clarified if the participants were looking at the values correctly for the second question, and the second question is an estimation of A/B, where $A < B$. Accuracy was measured by $\log_2(|\text{True value} - \text{Estimated value}| + 1/8)$, using the trimmed averages when bootstrapping confidence intervals. This measure of error was used to assess which type of chart was better and provided recommendations for how to display information. A replication performed by Heer & Bostock (2010a) used Amazon’s Mechanical Turk, which also found similar recommended chart features using a crowdsourcing platform. Many other studies tend to use similar types of questions when tasking participants to make numerical estimations.

When designing studies for numerical estimation, the choice of A/B is an important consideration. Many researchers seemingly choose ratios in an arbitrary fashion or follow previously established work (e.g., Zacks et al. (1998) used the same ratios as Melody Carswell et al. (1991)). Examples of ratios used in several studies are provided in Figure 1.12. However, the magnitude of stimuli and ratios are suggested to be a contributing factor to the amount of error in estimations (Cleveland & McGill, 1984a; Zacks et al., 1998). Additionally, participants may display an anchoring effect if the values are close to certain increments (e.g., estimating 55% as 50%). This suggests that the selection of ratios for estimation should cover a wide range of possible ratios.

Several calculations for error can be considered when estimating accuracy. The magnitude of error is calculated as $|\text{Guess} - \text{True value}|$, which is how far away a participant's estimate is from the target value. Removing the absolute value gives directional error, which allows for estimates to indicate bias of overestimation or underestimation. Since ratio estimations tend to be exponential (Veit, 1978), it is also common to use the logarithmic transformation, $\log|\text{Error}|$, to address non-normality for the magnitude of error. When reporting which chart type is more accurate in the extraction of information, the calculation of error needs to be clearly defined.

The type of estimation task can influence how participants respond with their estimates. Psychophysics has demonstrated that asking participants to estimate ratios elicits a different mental process than when asking participants to estimate differences (Birnbaum & Mellers, 1978; Veit, 1978). Estimations of ratios tend to fall on an exponential scale, where larger differences in stimuli result in smaller differences of estimated ratios. This makes $\log|\text{Error}|$ a natural choice for measuring ratio estimation tasks, similar to how Cleveland & McGill (1984a) measured error. Measurements of differences tend to be estimated linearly, where estimated differences are the same regardless of differences in stimuli. Modern statistical methodology is able to accommodate these linear and exponential relationships, as long as the estimation task is consistent for each trial.



(a) Cleveland & McGill (1984a)

(b) Croxton & Stein (1932a)

(c) Zacks et al. (1998), Experiment 1

(d) Zacks et al. (1998), Experiment 3

Figure 1.12: Ratios used by several studies

1.7 Research Gaps and Study Motivation

It is generally advised to avoid using 3D data visualizations whenever possible, although previous studies often contradict this statement through mixed findings. The main contributing factor tends to be the role of the third dimension. When the third dimension is used for depth, 2D charts are generally preferred for better numerical extractions and quicker completion times. On the contrary, using the third dimension for displaying information can provide better estimates than 2D alternatives. In addition, 3D charts tend to be chosen for their ability to improve, or attempt to improve, the memorability of the data. This makes previous research inconclusive on a superior chart dimensionality.

Despite extensive study of graphical dimensionality, these findings are almost exclusively based on 2D projections presented on paper or digital displays. As a result, current recommendations may not generalize to contexts in which the third dimension is physically realized rather than visually implied. Physical charts allow for natural interactions and promote additional engagement with the data structures. However, current literature provides little guidance on the construction or effectiveness of physical charts.

In this research, we explore the the physical dimensionality of 2D and 3D statistical graphics. In Chapter 1, we explore the dimensionality of bar charts using a pilot study that partially replicates the work of Cleveland & McGill (1984a). We then expand the study in Chapter 2 as an experiential learning project for an undergraduate introductory statistics course and collect information on how students reflected on various parts of the scientific process. Chapter 3 takes the ideas of Chapter 1 and Chapter 2 and replaces bar charts with heatmaps. Collectively, this research provides the initial large-scale studies into physical statistical graphics, resulting in recommendations for the creation and use of tangible data visualizations.

Chapter 2

Evaluating Perceptual Judgements on 3D Printed Bar Charts

2.1 Introduction

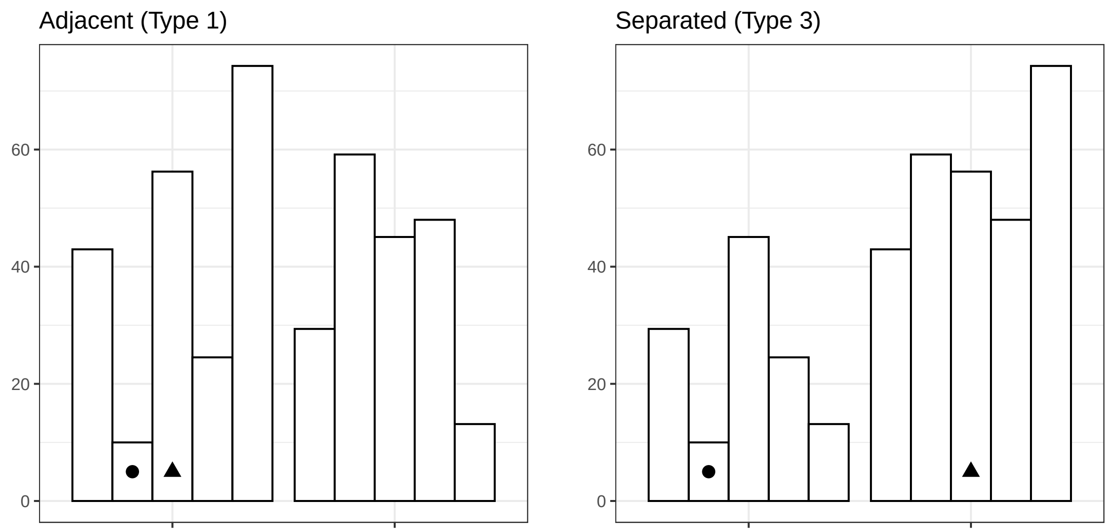
Good communication requires both that the information be transmitted correctly and that the intended recipient be able to decode and understand the transmitted information accurately. In order to communicate effectively, we must use graphical forms that accurately convey information relevant to the task in question. In many cases, this means we must understand how accurately people can read quantitative information off of charts. While accuracy is not the only quantity of interest in graphical investigations (Hullman et al., 2019), it is an important factor in assessing the utility of many different data graphics.

The accuracy of graphical forms has been studied for almost a century (Croxton & Stein, 1932b; Croxton & Stryker, 1927b; Eells, 1926a; von Huhn, 1927); as new ways of representing information evolve, we must revisit old studies to determine whether these representations have the same limitations as previous versions. This is particularly true in areas like graphics which are affected by the immense technological innovation in hardware and software which has taken place since the early 1990s.

2.1.1 Elementary Graphical Tasks

Cleveland & McGill (1984b) established the comparative accuracy of different "elementary perceptual tasks" (EPTs). Elementary Perceptual Tasks, according to these experiments, include assessing graphical elements such as position along a common scale, length, angle, and volume, and estimating the corresponding numerical value of these representations. The study relied entirely on estimation accuracy, which may not always be relevant when extracting information from graphs. For example, estimation is less relevant when ordering values by size. As a result of the Cleveland and McGill (1984b) study, it is possible to assemble an ordering of perceptual accuracy for the elements of length, position, and angle. Heer & Bostock (2010b) replicated some parts of Cleveland & McGill (1984b) in an online setting using Mechanical Turk, a platform for crowdsourcing human tasks that has been used for (among other things) assembling machine learning datasets. This replication study largely validated the results of the original study while demonstrating the utility of the Mechanical Turk platform for graphical testing.

The first experiment in Cleveland & McGill (1984b) (the position-length experiment), used five types of bar charts: two types of grouped bar charts and three types of stacked bar charts. Each chart had two bars marked for comparison; participants were asked to determine which bar was smaller and give the perceived ratio of the smaller bar to the larger bar. Figure 2.1 shows the two types of grouped bar charts. We are primarily interested in the grouped bar charts (in part because 3D printing is not yet inexpensive enough to make moderate-scale stacked bar chart experiments viable), which consisted of two comparison bars which were either adjacent or in separate groups. These grouped bar charts will be referenced as adjacent and separated graph types in this paper, respectively.



(a) Type 1 (b) Type 3

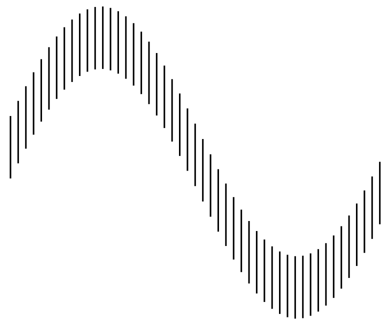
Figure 2.1: Cleveland & McGill (1984b) used two different types of grouped bar charts: comparisons between adjacent bars, and comparisons between separated bars. It is widely acknowledged that comparisons between separated bars (Type 3 comparisons, in Cleveland & McGill's terminology) are more difficult and error-prone.

2.1.2 3D Graphical Perception

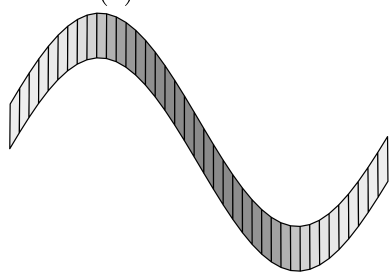
Chart perception is affected by the human visual system's implicit assumption that visual stimuli are three-dimensional; after all, most of the visual input we process does come from a three-dimensional world, but charts are artificial. In addition, these artificial stimuli are typically created and presented in two dimensions. The artificial scenes we create for data visualization have been documented to cause problems: for instance, the line-width illusion (Day & Stecher, 1991; Hofmann & Vendettuoli, 2013; VanderPlas & Hofmann, 2015) has been attributed to implicit 3D perception of 2D stimuli and can affect perception of error bands, candlestick plots, hammock plots, and Sankey diagrams. The line width illusion causes lines or error bands which are actually the same length to appear shorter at points with large amounts of curvature; a simple example of the classic sine illusion (one version of the line width illusion) is shown in Figure 2.2. While this illusion is problematic in two dimensions, when the depth cues are actually present in a three-dimensional situation, the same implicit visual heuristics contribute to accurate size perception (as in the middle panel of Figure 2.2). Thus, there is reason to think that perceptual accuracy may be dependent on the realism of the visual display relative to the training of the visual system and the heuristics which are activated in order to make sense of the display.

The use of 3D graphics have been explored in multiple studies, with mixed results. Fisher et al. (1997b) explored user preference for 2D or 3D charts and found that subjects tended to prefer simpler 2D graphs when tasked with extracting information. Barfield & Robless (1989b) compared 2D and 3D graphs presented on paper and on computers, showing that the accuracy of subject answers depended on their skill level: novice subjects were more accurate with 2D paper graphs, while experienced managers were more accurate with 3D computer graphs. For both experience levels, participants were more confident in their answers when using 2D graphs. There are instances where 2D graphs perform better than 3D graphs, but there are times where 3D graphs may provide a more natural way to encode information, at least in theory. For instance, when X and Y are used to represent spatial dimensions, it may be preferable to use a 3D chart to convey numerical information instead of using color, which is perceived much less accurately. Brath (2014) highlights the intrinsic attributes of 3D graphs and the benefits when used appropriately with other 3D elements such as lighting and correct portrayal of data attributes. Nevertheless, it is extremely common in visualization literature to recommend that 3D plots be avoided at all costs (Morgan-Wall, 2019; Vanderplas et al., 2020b; Wilke, 2020).

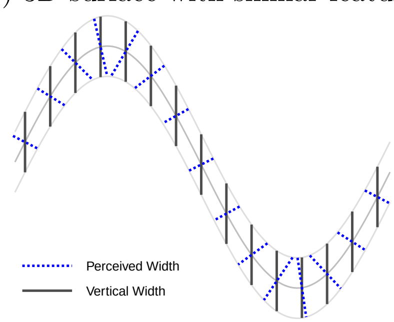
There is thus good reason to be wary of the use of three dimensions where only two are necessary to convey data (Huff, 1954). However, the situation is different



(a) Sine Illusion



(b) 3D surface with similar features



(c) Perceived vs. actual length

Figure 2.2: An illustration of the sine illusion (VanderPlas & Hofmann, 2015), also known as the line-width illusion. All vertical lines are the same length, but the lines in the middle of the curve appear to be much shorter. The illusion results when implicit perceptual corrections useful for perceiving the size of objects with depth are applied to 2D stimuli with no actual depth. As 3D heuristics occasionally cause inaccurate communication when applied to 2D objects, it is reasonable to think that there might be some situations where charts making use of a realistic third dimension might be perceived more accurately than their 2D equivalents.

now than it was in 1984 when Cleveland & McGill published their seminal work; it has even changed since Heer's replication study (2010b) in 2010.

Computer graphics and rendering technology have developed quickly, along with the hardware necessary to support these software developments. As a result, we have much more natural virtual rendering of 3D objects, and we can also print graphics in three dimensions using mass-market, relatively inexpensive 3D printers, moving artificial charts into a more natural, physical setting. As a result, it is reasonable to reconsider the use of 3D charts, not only because of new technological developments, but also because 3D printed charts could eventually provide the opportunity to make data graphics accessible to those with limited or absent vision (Fleet, 2023).

In this paper, we discuss a study designed to examine Cleveland & McGill's experiments on grouped bar charts using modern graphics in two and three dimensions. This study lays the groundwork for additional empirical studies on the use of 3D graphics, rendered and 3D printed, for visualizing complex data. In order to explore perception of fully 3D graphics, it is prudent to start with the simplest possible 3D graphic: one in which the third dimension is not necessary, so that we can easily compare to two-dimensional representations without loss of information. In the next section we provide details about design and execution of our experiment, including the process of replicating stimuli from Cleveland & McGill (1984b) in order to create 2D, 3D projected, 3D rendered, and 3D printed bar graphs. The next section presents the results, and we conclude the paper by discussing this experiment in the context of existing work on the perception of 2D and 3D graphical elements.

Table 2.1: Values used in the experiment to make ratio comparisons, sorted by ratio. The ID label corresponds to the file reference number.

Bar	ID: 01	ID: 06	ID: 05	ID: 04	ID: 02	ID: 03	ID: 09
Smaller	10.0	14.7	14.7	12.1	10.0	12.1	26.1
Larger	56.2	56.2	38.3	26.1	17.8	17.8	31.6
Ratio (%)	17.8	26.2	38.4	46.4	56.2	68.0	82.6

2.2 Methods

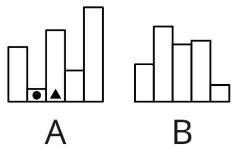
Our study is designed to replicate and expand upon the position-length experiment from Cleveland & McGill as closely as possible, with the hope of being able to integrate this study with previously reported results in a consistent manner. In this section, we discuss the replication process and the design of our version of this experiment.

2.2.1 Replicating Cleveland and McGill

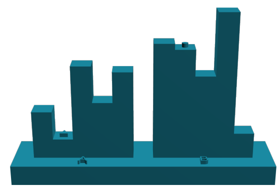
The first step of replicating the position-length experiment was to determine the heights of the bars that participants use for comparisons. These values for the bar heights are linear on a log scale and are given by

$$s_i = 10 \cdot 10^{(i-1)/12}, \qquad i = 1, ..., 10$$

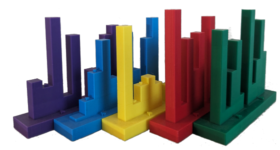
Each graph presents two bars from the values given above where the participants are asked to judge the ratio of the smaller bar to the larger bar. The ratio of bars used by Cleveland and McGill were 17.8, 26.1, 38.3, 46.4 (twice), 56.2, 68.1 (twice), and 82.5 (twice). The exact numeric comparisons were not disclosed, but the comparison values used in our study were subjected to the constraints of having the same ratio values and that no value was used more than twice.



(a) 2D bar chart



(b) 3D digital bar chart



(c) 3D-printed bar chart

Figure 2.3: Two dimensional, 3D digital rendering, and 3D-printed charts used in this study.

In each graph, there are two sets of five bars each, and two of the bars are marked for identification with a circle and triangle, as in Figure 2.1. Cleveland and McGill did not specify the random process for the heights of the eight other bars, so we used a scaled Beta distribution to provide random structure around our fixed values. Code to reproduce the data generation process, data underlying the plots used in this study, the rendered plots and STL files, anonymized user data, and analysis code can found be at `https://github.com/TWiedRW/2023-JDS-3dcharts`.

2.2.2 Stimuli Construction

The graphs share a common layout across all formats, where two groupings of five bars are identified by “A” and “B”, respectively, and circles and triangles are used to identify the bars participants should compare. Example graphs are shown in Figure 2.3. There are some graphical elements that cannot be easily portrayed via 3D printing. For this reason, all graph types do not have axes, grid lines, or floating titles.

The ggplot2 (Wickham, 2016) package was utilized to create the 2D bar charts. The scale axis was removed, leaving only the bars and a bar grouping identifier. The bars used for comparisons had the identifying mark at a height of 5 out of 100 for the 2D plots, and the 3D plots had the identifying marks on top of the bars.

The 3D renderings and 3D printed charts were both created using OpenSCAD (Kintel, 2023), which creates STL files from markup describing the object's geometric composition. Charts were composed of a platform, with raised text labels centered in front of the A and B groups of bars. Bars were created by inserting bar heights into the OpenSCAD template using R, and then raised circle and triangle markers were positioned on top of the bars to be compared to provide a tactile and visual indicator. In addition, an ID code was engraved into the bottom of the platform to uniquely identify each object; this allowed the researchers to ensure that each kit contained the correct charts throughout the experiment while minimizing the risk of stickers or other identifiers becoming detached from the charts. As printed, the base of each chart was 13cm x 3cm x 1cm, with the highest bar rising 9.5cm above the chart base. Raised letters and shapes were 2mm above the base or bar, respectively. The color of the filament used to print each chart corresponded to the estimation ratio (though this was not obvious to participants), allowing the researchers to quickly determine that a kit of 3D charts contained no ratio duplication. Colors were randomly assigned to ratios so that participants could not gain information about the ratio from any perceived ordering of the colors.

Digital renderings of the generated STL files were created using RGL (Murdoch & Adler, 2023), which integrates into Shiny (Chang et al., 2023) using the WebGL (Mozilla Foundation, 2023) extension. The RGL rendering of the STL file was initially angled to correspond to the perspective of default 3D bar charts present in Microsoft Excel, but in the WebGL interactive environment, participants could rotate the charts as they pleased to create a useful estimate of the bar ratio. Rendered charts were colored to correspond with the filament color used to print the physical chart for consistency. The default `rgl` lighting was replaced with three lights located in fixed positions around the rendered figure. The lights were positioned so that one was behind the rendered figure, another in front of the figure, and one light below the figure. This arrangement provided a consistent visual experience and minimized shadows that prevented visual estimation of the bars.

2.2.3 Experiment Design

A diagram of the experimental design is provided in Figure 2.4. The study is set up as a modified randomized incomplete block design, comprised of 3 blocks (presentation mode) and 7 bar ratios (5 are selected for each participant). At each combination of ratio and presentation type, the comparison type, adjacent (Type 1) or separated (Type 3), is randomly selected. This process was used to create 21 kits of 5 3D printed charts and 10 virtual charts. There is thus a slight deviation from a completely randomized incomplete block design, in that all possible combinations of ratios, presentation modes, and comparison types are not present in our design. However, this approach provided the best experimental design given the constraints of managing physical stimuli; it took nearly a month of continuous 3D printing to create all of the charts for the kits that we assembled. A dataset containing the kit composition breakdown is available in the github repository for this project.

2.2.4 Participant Recruitment

One interesting facet of Cleveland & McGill (1984b) is the participant recruitment methodology: "For each experiment the subjects fell into two categories: (1) a group of females, mostly housewives, without substantial technical experience; (2) a mixture of males and females with substantial technical training and working in technical jobs. Most of the subjects in the position-length experiment participated in the position-angle experiment; in all cases repeat subjects judged the position-angle graphs first." It seems likely that the authors

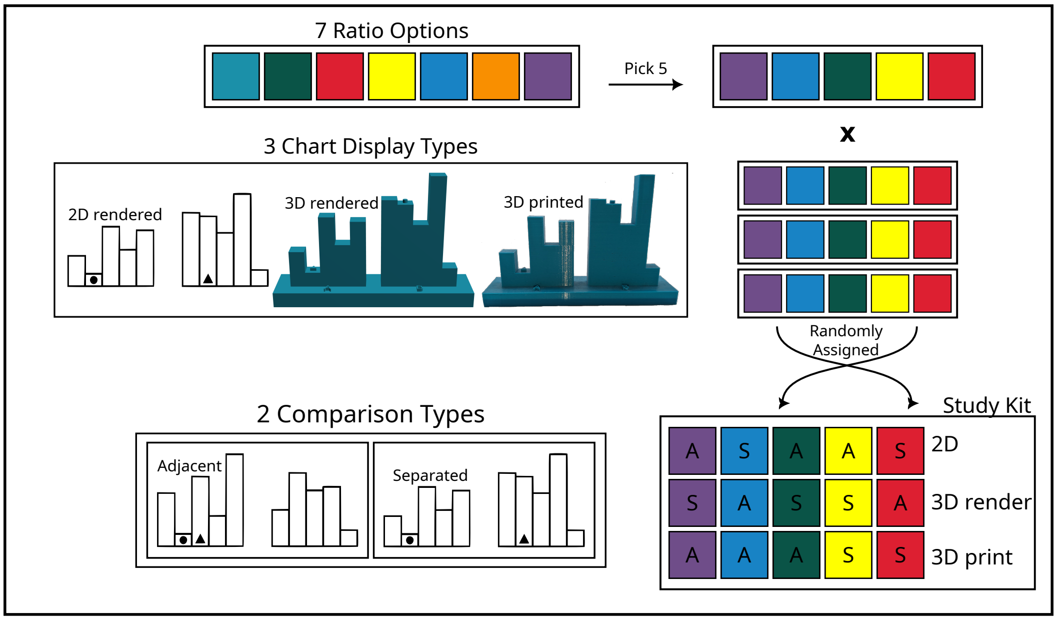


Figure 2.4: A graphical representation of the study design. Only five of the seven ratios were used in each kit; at least one of the smallest or largest ratios were included along with 4 other charts; each selected ratio was displayed in all 3 mediums. For each medium × ratio combination, the comparison type (separated or adjacent) was randomly determined. A total of 21 kits of 3D printed charts were created to include all combinations of the five ratios.

recruited individuals within their respective departments as well as their wives. In order to replicate this atypical participant recruitment method in the modern era, members of the UNL Statistics department and their spouses, partners, and roommates were asked to participate in our study. This replicates the spirit of the recruitment method without the implicit assumptions that graduate students and professors are male, heterosexual, and have unemployed spouses. A total of 38 participants completed our study; demographics are shown in Figure 2.5.

While the results of any sample with recruitment methodology like this are not generalizable to the public, running our initial study on a comparable population to that of the initial study serves a purpose: other replications, such as Heer & Bostock (2010b), used online samples of paid participants, who likely have less

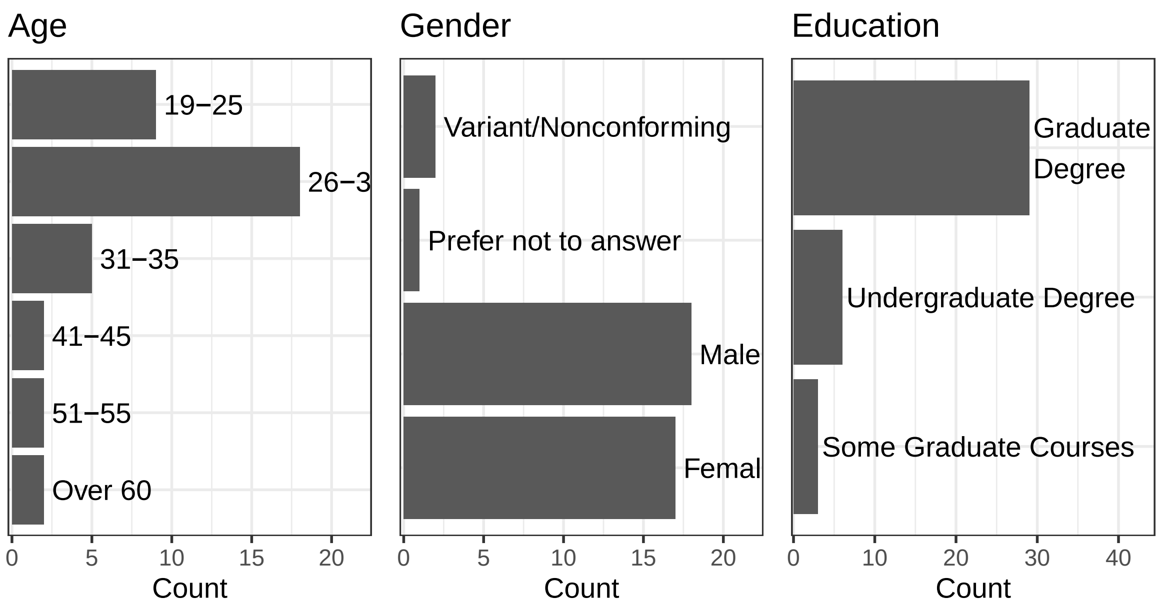


Figure 2.5: Demographic characteristics of participants in the study.

experience using charts and graphics and making numerical estimations than those in a statistics department (and individuals who cohabitate with them). We have every intention of running this experiment in other populations, but the convenience of online sampling is not compatible with physical objects such as our 3D printed charts, so we will have to recruit in-person participants. Thus, an initial study in a population comparable to Cleveland & McGill (1984b) is a reasonable first step into testing 3D charts.

2.2.5 Data Collection

A Shiny applet was used for data collection, along with the provided kit of 3D printed charts. Participants provided informed consent through the applet, and then were asked for demographic information (age, gender, education level). Then, participants were shown a “practice” page which allowed them to experiment with the data collection interface and practice estimating the ratio between the bars, as shown in Figure 2.6. This practice interface did not provide any numeric feedback

Figure 2.6: Screenshot of Shiny application practice screen. Three 2D bar charts with different ratios were provided, along with sliders indicating the correct proportion. Participants could practice with the sliders and preview the questions that would be asked as part of the task.

as to participant correctness, but was merely intended to familiarize the participants with the questions which would be asked as well as the process of estimating the ratio between the two bars.

After the practice screen, participants were asked to provide the kit ID. Participants were also instructed that when indicated, they should choose a 3D chart from the kit and select the ID code of the chart (inscribed on the bottom) from the drop-down menu before completing the estimation task. Participants were also instructed to make quick judgements for each graph and not to measure or estimate ratios using physical objects.

Each graph (or prompt, in the case of 3D printed charts) in the applet had two corresponding questions for participants to answer: first, participants were to identify the smaller bar by shape, and then, participants were to estimate the ratio of the size of the smaller bar to the size of the larger bar, as shown in Figure 2.7. It

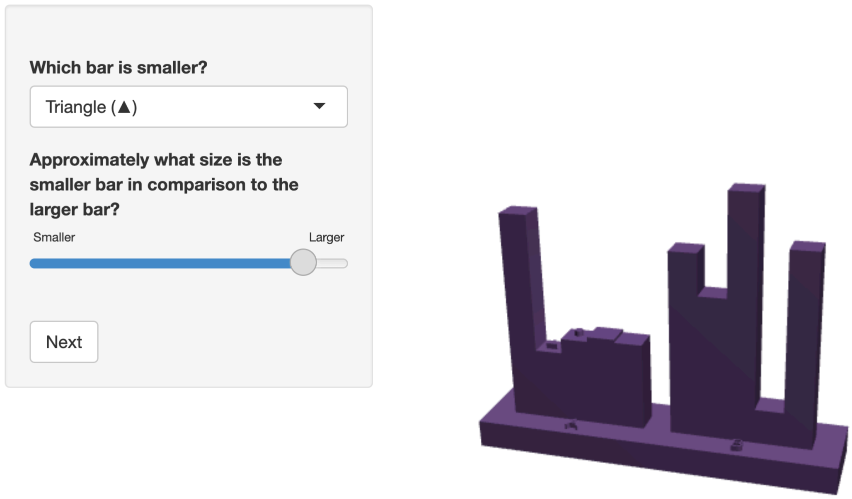


Figure 2.7: Screenshot of the applet collecting data for a 3D rendered chart task. Participants were asked to select which bar (circle or triangle) was smaller, and then to estimate the ratio of the smaller bar to the larger bar.

should be noted that we chose to use a slider without numeric identifiers or tick marks for entering ratio estimates - in part, this was intended to alleviate rounding effects which occur when participants enter numbers on their own or get numeric feedback from slider entries (Maineri et al., 2021; Ruud et al., 2014). Another benefit of this modification is that it should reduce participants' cognitive load, allowing them to focus on the ratio between bars rather than the numerical translation process. This differs from Cleveland & McGill (1984b), but maintains the spirit of the study while using methods which were not available for data entry at the time.

2.3 Results

The participants provided a total of 495 responses. While each participant was provided with 15 trials, 6 participants partially completed the experiment; one participant did not advance beyond the demographic page. All responses that incorrectly identified the smaller bar ($n = 18$) were removed from the study before analysis; this serves as a basic attention check and leaves 477 valid responses. In addition, one response was removed because the participant did not enter a valid ID for a 3D printed chart.

2.3.1 Midmeans of Log Absolute Errors

Cleveland & McGill (1984b) used

$$\log_2(|\text{Judged Percent} - \text{True Percent}| + 1/8)$$

to measure accuracy of their participant's responses. In their study, log base 2 seemed appropriate due to "average relative errors changing by factors less than 10." They also added 1/8 to prevent distortions when the errors were close to zero. Heer & Bostock (2010b) followed the same analysis method, replicating many (but not all) of the results presented in the original paper.

Figure 2.8 shows the midmeans of the log absolute errors compared to the true ratio of the bars for each graph type and comparison type. The results suggest that the log absolute errors increase for greater differences between the smaller and larger bars, but are consistent across the graph types.

This is somewhat different than the results in Cleveland & McGill (1984b) and Heer & Bostock (2010b); in both cases the midmean log absolute errors increased until about 55% of the true proportional difference and then decreased. It is possible that this difference is due to the fact that we did not require an explicit numerical estimate of the ratio but instead asked participants to indicate the ratio on a slider (which is essentially a number line). It is possible that this change explains the lack of a difference between chart modalities we see, but there are other

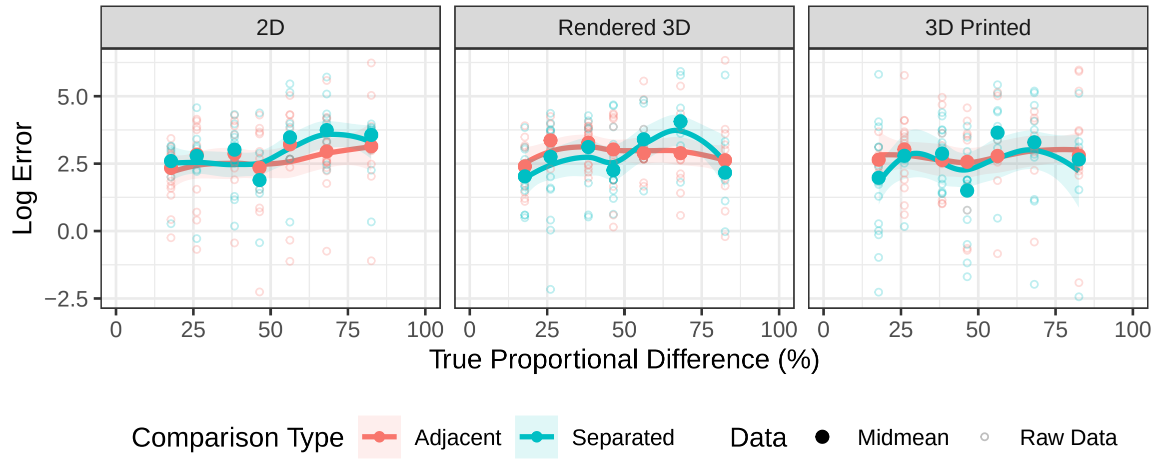


Figure 2.8: Midmeans and observed values of log absolute errors for the true ratio of bars. Summary lines are computed from raw data using a loess smooth.

potential explanations as well. One interesting facet of this data is that there is a consistent reduction in log error when the true proportional difference is near 50%. This may be because of implicit anchoring - we can typically bisect a span relatively accurately, which means that it might be expected that when the true proportion is near 50%, we would both notice that and be able to accurately transfer that information to the slider. This pattern is more noticable in separated (Type 3) comparisons, which also makes sense - as these comparisons are known to be less accurate, a locally minimal error near 50% with increased error relative to adjacent (Type 1) comparisons could easily explain the exaggerated trends we see in Figure 2.8.

2.3.2 Generalized Additive Model

In addition to replicating the (primarily graphical) analysis of participant errors, we also took a more statistical approach and fitted a generalized additive model (Hastie, 2017) that accounts for participant variation as well as the effect of comparison type, graph type, and ratio.

We define the following effects:

- R, the numerical ratio of A/B where A is the length of the smaller bar and B is the length of the larger bar. There are $i = 1, ..., 7$ levels of R in this study, but we will treat R as a numeric variable throughout this model.

- D_j, the display mode, where $j = 1, 2, 3$ correspond to 2D, 3D rendered, and 3D printed charts respectively.

- T_k, the type of comparison, where $k = 1, 2$ correspond to adjacent (Type 1) and separated (Type 3) comparisons, respectively

- P_l, a random effect for participant which describes the overall skill of the participant at visual estimation of ratios.

Generalized additive models depend on a smooth function $s() : \mathbb{R} \to \mathbb{R}$ which is applied to the independent variables; by default, the smooth function is created using thin-plate regression splines. In place of an interaction term when a categorical variable is involved, separate smooths are fit for each level of the categorical variable; we will indicate this by subscripting the smooth function with the categorical variable index, so that $s_1^T(R)$ indicates the smooth over numerical variable R for the 1st level of the categorical variable T, that is, the smooth function over R for adjacent comparisons.

Then the formal statistical model is as follows:

$$y = s_j^D(R) + s_k^T(R) + D_j + T_k + P_l + \epsilon$$

where

- $y = \log_2(|\text{Judged Percent}_{ijkl} - \text{True Percent}_{ij}| + 1/8)$

- $s_j^D(R)$ is a thin-plate spline function for R accounting for the display type
- $s_k^T(R)$ is a thin-plate spline function for R accounting for the comparison type
- D_j is the fixed effect of the display type (e.g. an intercept term)
- T_k is the fixed effect of the k^{th} comparison type
- P_l is the random effect of the l^{th} participant
- ϵ is the random error

Table 2.2: Parametric coefficients in gam model.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	2.6862	0.1780	15.0931	0.0000
plot3dDigital	0.1357	0.1409	0.9626	0.3363
plot3dPrint	-0.0563	0.1415	-0.3980	0.6908
typeAdjacent	0.0968	0.1197	0.8088	0.4191

Table 2.3: Approximate significance of smooth terms in gam model.

Smooth	edf	Ref.df	F	p-value
s(Ratio):plot2dDigital	1.0005	1.0009	2.4500	0.1183
s(Ratio):plot3dDigital	1.7934	2.2056	0.7146	0.4521
s(Ratio):plot3dPrint	0.0003	0.0005	0.0004	0.9996
s(Ratio):typeSeparated	4.2371	4.7338	4.2693	0.0018
s(Ratio):typeAdjacent	1.0007	1.0014	0.0059	0.9429
s(participant)	27.0449	34.0000	4.7272	0.0000

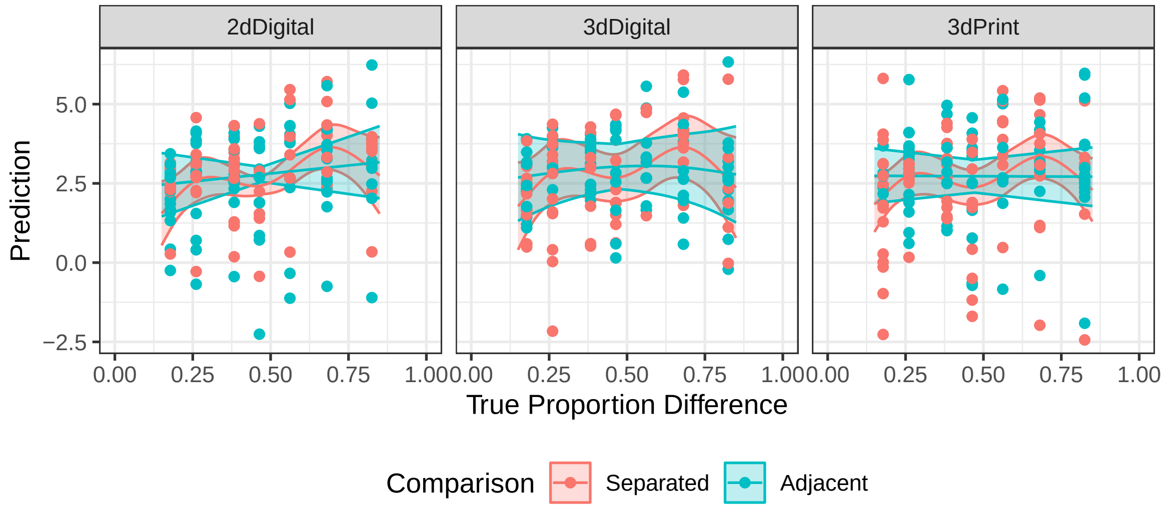


Figure 2.9: Predictions with standard errors for fixed effects in a generalized additive model with splines for ratio by comparison and ratio by display method. It is clear that the separated comparisons are easier to estimate when small, right around 50%, or large, and harder to estimate between these points. What is interesting is that no such trend is present for adjacent comparisons.

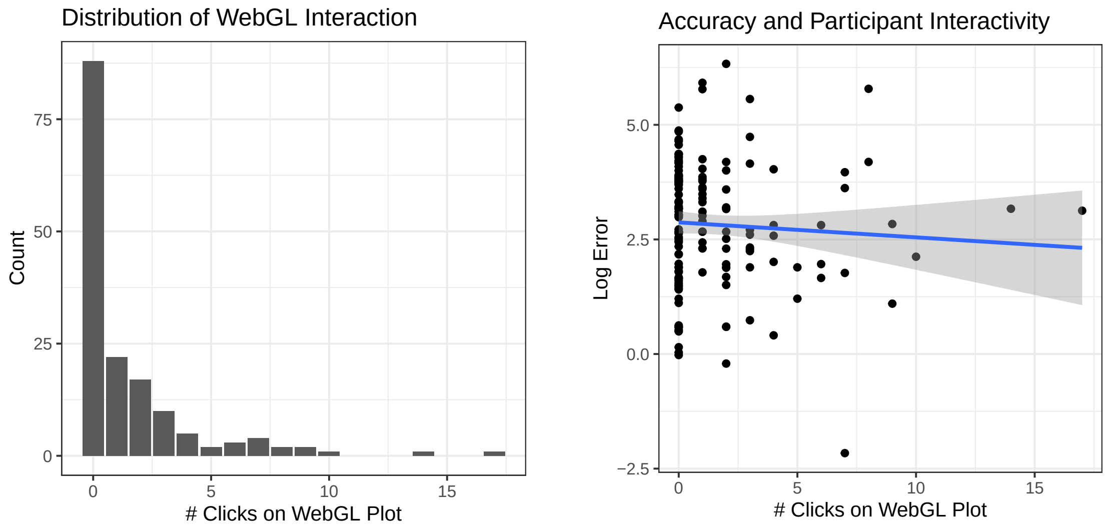
A generalized additive model with random effects for subject was fit using the `mgcv` package (Wood, 2003, 2004; Wood, 2017a). Spline terms by comparison type and display type were included for the ratio variable, to account for indications of nonlinearity from Figure 2.8; cumulative predictions for fixed effects are shown in Figure 2.9. It is clear that there are differences between the errors for separated comparisons and adjacent comparisons; adjacent comparisons seem to have roughly similar errors for all ratio values, while separated comparisons seem to have lower errors for points near 0, .5, and 1, with higher errors as the distance increases from one of these anchor points; this effect has been explored in Hollands & Dyre (2000). Residuals from the model indicate that there is some left skew in the tails, but this is not extreme; code for this analysis can be found in the supplemental materials on GitHub.

While we were initially concerned that this effect was due to the slider initial setting of 0.5, we did examine this hypothesis; only 42 responses (8.81%) had slider values at exactly 0.5, and all but 6 were for ratios on either side of 0.5 (.464, .562). This suggests that while a few participants did take the strategy of using the default value for the slider, they did so strategically, not moving the slider only when the ratio was perceived to be sufficiently close to 0.5. In addition, trials where participants utilized this strategy were evenly split between separated and adjacent comparisons, suggesting that this anchoring process is not the reason for the difference in spline fits for the different comparison types. It seems more likely that this effect is at least partially due instead to lower errors around the extremes of the ratio spectrum. A follow-up experiment with more ratios and in particular ratio values closer to suspected anchor points could explore this relationship in more depth - with only 7 different ratios, it is difficult to do more than note the appearance of an effect and speculate about the anchor points, particularly given that we have used spline models with 5 knots to fit the smooth.

Heer & Bostock (2010b) provided a zip file containing data and code for their paper, but this link has not been updated and is no longer active, so it is not possible to fit a similar model to their data for comparison purposes; if we could get access to their data, this would allow for investigating whether the differences identified in our model were an artifact of the measurement method (e.g. slider vs. numerical entry) or if this effect was present in previous studies and can now be identified because of more advanced statistical modeling techniques.

2.3.3 Interactivity with 3D Rendered Charts

When 3D rendered plots were shown, we recorded the number of user interactions (clicks, rotations) with the WebGL plot. Figure 2.10 shows the distribution of number of clicks over all 3D rendered trials in the experiment, as well as the number of clicks relative to the accuracy measure



(a) Histogram of clicks

(b) Clicks vs. log error

Figure 2.10: Participant clicks on the WebGL interface (left) in order to interact with the plot. Most participants did not interact with the WebGL interface, perhaps because they did not realize that interacting was an option, but some participants utilized the interactivity heavily. (Right) Participant interactivity was not associated with increased accuracy.

$\log_2(|\text{Judged Percent} - \text{True Percent}| + 1/8)$. It does not appear that participants who interacted with the WebGL rendering were more accurate than their peers who did not.

2.4 Discussion and Future Work

Previous work in 3D graphics would suggest that the errors for the 3D graphs would be larger than the errors for the 2D graphs. While we did not find any significant results indicating that 3D graphs are read less accurately, there are two possibilities that might account for this discrepancy.

The first potential explanation is that this study is underpowered - the effect size is small, and our 38 participants were insufficient; the original study included 51 participants, which is slightly larger. However, we should note that the analysis methods used in Heer & Bostock (2010b) and Cleveland & McGill (1984b) do not provide numerical tests of statistical significance, instead defaulting to graphical displays which include confidence intervals.

The second possibility is more interesting: we examined 3D charts using rendered 3D graphs and 3D printed charts; both of these options allow for participants to interact with the chart, rotating it, and generally perceiving it as one might perceive any other 3D, real, object. This is a far cry from the 3D perspective charts in the original study, which have a fixed angle and perspective and are thus not equivalent to our 3D charts. Future studies should include an additional fixed 3D perspective bar chart, which will at least enable us to examine whether modern 3D rendering environments allow for more accurate conclusions than fixed 3D perspectives. Future iterations of this study will include “traditional” 3D graphs created by Microsoft Excel (that is, graphs with a fixed 3D perspective rendered in 2D). This option will allow us to examine fixed perspective 3D plots compared to 3D renderings and 2D plots; it will also enable online data collection in addition to the in-person data collection used in this experiment.

Another interesting aspect of our study is that the method used to record participant estimates is different from the method used in the original study as well as Heer & Bostock’s replication study. A method similar to our slider input, marking position on a line, was used in Spence (1990), but Spence asked participants to estimate $A/(A + B)$, where we asked participants to estimate A/B; thus, our results are still not directly comparable to previous studies. We expect that the specific ratio estimated would also have an effect on observed participant errors.

The slider method for input of ratio estimates should be easier for participants, as it does not require explicit transformation to the numerical domain. What is clear is that it would be beneficial to assess the impact of the measurement method on participant errors directly, so that the results of these different studies might be explained and interpreted with regard both to the stimuli used and the measurement method employed in the experiment. Future iterations of this experiment will likely address this estimation difference; such modifications in experimental design are relatively straightforward in Shiny and will provide useful insight into the design of future experiments evaluating the perception of statistical graphics.

2.4.1 Supplemental Material

Stimuli, code, and data for this experiment are provided at `https://github.com/TWiedRW/2023-JDS-3dcharts`.

Chapter 3

Experiential Learning (PLACEHOLDER)

3.1 Introduction

The education of statistics emphasizes the use of real data and its application in answering research questions. Students in introductory statistics courses are mainly exposed to elementary methods and textbook examples that demonstrate the application of these methods, which is used, in part, because of the emphasis on teaching students to think statistically using real data (Carver et al., 2016). Many textbooks, such as Tintle et al. (2021), use scenarios and ask students to perform the corresponding inferential test. This process is then repeated over the course material without much deviation in the instructional method. In some cases, however, students can participate and benefit from well-designed classroom experiments (McGowan, 2011). This process can expose students to concepts such as randomization and let students see the specifics of experimental design through their participation. Loy (2021) has demonstrated that student participants often recalled their experiment in later concepts, showing some evidence that students can benefit from the hands-on experience.

In addition to using experiential learning broadly, a key aspect in the statistics classroom is teaching students to interpret data through graphs. Nearly 40 years ago, Cleveland & McGill (1984a) began the process of establishing good practices for making graphs, although characteristics of creating graphs have been studied for nearly a century (Vanderplas et al., 2020a). While their findings have been replicated (Heer & Bostock, 2010a), many areas in data visualization remain underdeveloped that can benefit from the framework used by Cleveland & McGill (1984a). One such area is the projections of 3D data visualizations. The current mantra is to avoid 3D graphs when possible and studies around the 1990s seem to provide some valid skepticism of their use (Barfield & Robless, 1989a; Fisher et al., 1997a). Barfield & Robless (1989a) showed that participants were sometimes more accurate using 3D graphs than 2D graphs, depending on the participant's experience level. However, participants were generally more confident with their answers for 2D graphs. Fisher et al. (1997a) observed a similar preference for 2D graphs over 3D graphs when extracting information, but found no preference for visual appeal for the type of graph. While these results provided valid skepticism toward the use of 3D graphs, the results are generalized to the projections of the 3D graphs. The area of "true" 3D graphs is largely unexplored, but advances in technology allow for easier access to explore the 3D projections of these graphs. Kraus et al. (2020) explored the 3D projections with the use of virtual reality and found that participants were more accurate at extracting values from 3D heatmaps at the expense of needing more time than 2D alternatives.

Because the use of "true" 3D graphs remains largely unexplored, it provides a unique opportunity to be used as an experiential learning opportunity for statistics students. Not only can students benefit from the exposure to different graph types, including "true" 3D graphs, but they can also experience a more authentic method of teaching, which is more likely to be beneficial to reinforce statistical ways of thinking.

In this paper, we used an experiential learning project that employed different graph types, including "true" 3D graphs, in an introductory statistics classroom environment and described its potential application as an educational tool. We presented students with a series of modules where they first participated in our experiment on different graph projections, followed by acquiring knowledge on the purpose of the experiment by reading an extended abstract and watching a presentation of a pilot study of the same experiment.

3.2 Methods

3.2.1 Overview and Motivation

We introduced students enrolled in Introduction to Statistics (STAT 218) at the University of Nebraska-Lincoln to a graphics project that contained an experiment and progressively revealed components that illustrate the experiment's research objectives. The project started by providing students with minimal information about the research objectives before revealing the scope of the experiment through an extended abstract and presentation. The goal of the graphics project was to observe how students think statistically about experiments as both participants and research consumers. Students were provided with mostly open-ended prompts throughout the graphics project, which allowed us to freely explore common themes in student responses without the limitation of preset choices.

3.2.2 Experiential Learning

The graphics project was split into two main components regarding the interaction of students with the material. In this section, we will discuss student interactions with the experiment. Students were first presented with a series of four modules presented from the role of research participation. These modules contained the informed consent form, pre-experiment reflection, experiment participation, and post-experiment reflection. Within the informed consent module, students were informed that their data would be anonymized and that the experiment was exempt from the institutional review board (Project ID: 22579). While all students were given the option to participate in the graphics project, we were only able to collect responses when informed consent was obtained and if the student was 19 years of age or older. In the pre-experiment reflection, students were asked to write a paragraph about how they think the process of scientific investigation looks from the perspective of researchers and the general public. The experiment participation module asked students to paste the code generated upon completion of the graphics experiment, which is detailed in the next section. The generated code serves as a basic check to indicate whether or not students fully participated in the experiment. For the post-experiment reflection, students were asked five questions about the purpose of the experiment. These include questions on the hypotheses being tested, sources of error, variables of interest, and elements of experimental design.

After completing the experiment reflections, students moved to the reflection of the overall research objectives. Students were first directed to read an unpublished two-page extended abstract that we submitted as a contributed paper for the Symposium on Data Science & Statistics (SDSS). The extended abstract outlined the experiment's purpose and procedures, but not the results from our initial pilot study. After reading the extended abstract, students were asked to write a paragraph about what they found clearer about the experiment's purpose than when they were a participant. Finally, students were directed to watch a 12-minute pre-recorded presentation based on an abbreviated version given at SDSS (Wiederich, 2023). The video contained the same material as the extended abstract and included the results from our pilot study. The presentation reflection asked students four questions about the experiment and how information was presented differently than in the extended abstract.

Except for the informed consent and experiment participation modules, all student responses were open-ended. Each question and its corresponding module can be found in Table 1.

Instructors for STAT 218 were recruited for Summer 2023 and Fall 2023 to administer the graphics project into their classroom. The instructors were given the option of administering the project as coursework material or extra credit, along with the liberty of grading at their discretion. Instructors for the June-July Summer 2023 courses did not have the abstract or presentation modules.

3.2.3 Data Analysis

Since nearly all of the student responses to the project modules are open-ended, the analysis of the project modules is qualitative. We will selectively extract student responses that demonstrate variability and repetitive themes in their understanding of the graphics experiment. For all modules, except the Post-Experiment reflection, students are asked to comment on the experiment without an objectively correct response.

For paragraph responses in the Pre-Experiment and Abstract reflections, bigram plots are used as a graphical analysis to display word pairs after removing stop words (e.g., “the” and “and”). These word pairs help illustrate common themes that students wrote about in their longer prompts. In the Post-Experiment reflection, the prompts have objectively correct answers but would require a subjective aggregation to indicate the level of correctness. Instead, we opt to select at least one student response that we consider to be “most correct” and a couple of other responses that are either partially correct or entirely incorrect to illustrate the variability of the responses. The results of the graphics experiment are presented by Wiederich & VanderPlas (n.d.) as an extension of a larger study comparing the dimensionality and projections of 2D and 3D bar charts.

3.2.4 Graphics Experiment

3.2.4.1 Constructing Stimuli

Based on Cleveland and McGill's seminal work (1984a) on graphical perception, participants were presented with a series of bar graphs where two bars are marked with either a circle or a triangle. The heights of the bars were chosen from the following equation:

$$s_i = 10 \cdot 10^{(i-1)/12}, \qquad i = 1, 2, 3, ..., 10 \tag{3.1}$$

where s_i represents a value given an integer i as defined above. The values s_i from Equation 3.1 were then paired such that the ratio of the smaller value to the larger value yield the ratios of 0.178, 0.261, 0.383, 0.464, 0.562, 0.681, and 0.825. Each bar graph has two groupings of five bars. Following the Type 1 and Type 3 graphs from the position-length experiment by Cleveland & McGill (1984a), the value pairs for each ratio were either placed in the first grouping on the second and third bars (adjacent) or placed on the second bars in each grouping (separated).

To explore the effect of dimensionality and projection of the bar graphs, we introduced the following plot types: 2D digital, 3D digital (static), 3D digital (interactive), and 3D printed. There was no single software package that could create all four plot types, so we carefully constructed graphs from different software packages to be as similar as possible. The 2D digital plots were rendered with the `ggplot2` package (Wickham, 2016). Microsoft Excel® was used to render the 3D digital (static) plots. The 3D digital (interactive) and 3D printed plots were created with OpenSCAD® (Kintel, 2023), where the generated stereolithography (STL) files for the 3D digital (interactive) plots were rendered with the `rgl` package (Murdoch & Adler, 2023).

3.2.4.2 Experimental Design

With 56 treatment combinations, we opted to use an incomplete block design to provide participants with 15-20 graphs. Kits of graphs were constructed so that five of the seven ratios are equally represented, resulting in 21 different kits. Within each kit, all graph types appeared for each ratio and the comparison type was randomly assigned. A visual layout of the experiment is shown in Figure 1. All kits received a unique identifier and a set of instructions for accessing the experiment.

A Shiny application (Chang et al., 2023) was designed to administer the experiment. Students were directed to randomly select a kit of graphs and visit the Shiny application's website linked on the instructions. For students enrolled in the online sections of STAT 218, the website was provided by the instructor and they were prompted in the application to select that they were an online participant; selecting online participation resulted in the 3D-printed plots being removed from the set of graphs presented to the participant. After students provided a kit identifier (if applicable), students were presented with graphs in a randomized order. If the student marked that they were an online participant, the 3D-printed graphs were removed from their experiment lineup. Each graph asked the students to first identify the larger marked bar and then to guess the height of the smaller marked bar if the larger marked bar was 100 units tall using a slider widget. After completing the experiment, a completion code was generated for students to paste into the experiment participation module.

3.3 Results

3.3.1 Recruitment of Students and Instructors

We recruited 3 instructors for the summer and fall semesters in 2023. Each instructor offered the project as extra credit in their course and student participation was entirely voluntary. A total of 82 students participated in the project, and 9 students did not complete the entire project (Table 2).

3.3.2 Selected Responses from Experiment Participation

3.3.2.1 Pre-Experiment Reflection

In the first stage of the project, before participating in the experiment, students had limited information about the research objectives and we recorded how students thought about scientific research. Students were initially prompted with one question, "What do you think [the] process [of scientific investigation] looks like, from the perspective of a researcher, compared to what it looks like from the perspective of someone in the general public who is a consumer of scientific results?" The responses to this question allowed us to gather a broad understanding of what students initially thought about the research process. Students generally understood the purpose of scientific research by connecting the ideas of hypothesis testing and publishing results. Students wrote about scientific research starting from the place of a question, followed by conducting an experiment and relaying the results to the public. For example, the most common phrase groupings include variations of "scientific research", "data collection", and "peer review". A bigram plot of student responses to the Pre-Experiment reflection prompt is shown in Figure 2.

3.3.2.2 Post-Experiment Reflection

After participating in the experiment, students were provided prompts asking about the goals of the research objectives, which were to evaluate the errors of ratio judgments between 2D and 3D graph types under different comparison conditions in a randomized block design. Some students correctly identified parts of the questions asked in the post-experiment reflection, but most students typically missed some or most of what would be considered a correct answer.

We first asked students "What do you think the purpose of the experiment was?" For this question, a complete response would have included the comparisons of ratio judgments of the projections for 2D and 3D graphs. A student example of the correct response was "I think this experiment aimed to test if it was easier to compare two graphs in 2D or 3D." Another student missed the comparison of 2D to 3D graphs, but was mostly correct with their response "I think the purpose of this experiment was for the researcher to gather data on how people perceive, interpret, and understand 3D graphs." One student incorrectly thought that we were testing differences in demographics by responding "They could be trying to determine how different genders, ages, etc. perceive the sizes of the bars in the graph. Demographics could make a pretty significant difference."

We then asked students "What hypotheses might the experimenters have been testing?" A correct response would include testing the differences in errors of ratio judgments between graph types. One example of a student providing a correct response was given by stating "They might have been testing if a 2D model is easier to estimate its relative size to another when compared to a 3D model of it." Other students replied with statements that would not be able to be measured from the experiment, with one student responding "How taking Statistics 218 effects how you can compare two groups" and another student saying "That 2d is preferred over 3d. It cleans up the data presentation."

An important topic covered in STAT 218 is randomization, which we asked students with the prompt "What elements of experimental design, such as randomization or the use of a control group, do you think were present in the experiment? Why?" In the experiment, randomization was performed with the ordering and assignment of graphs into kits. One student perfectly described randomization by responding with "Randomization: The survey used an experimental design where in the survey there were different sets of 3D charts and maybe by a randomization process each participant was shown a different set of charts to see the differences in interpretations of the charts based on which set was assigned. Control Group: Since this survey aimed to only understand how participants interpret 3D charts without comparison to other chart types, then no control group was needed." Another student was partially correct with a response of "Randomization was used because the ever person got a different graph." Other students missed the use of randomization, with one student responding "random students in the stats class" and another saying "Randomization was not used because it was offered as an extra credit assignment in class."

3.3.3 Selected Responses from Research Reflections

In this set of reflections, students first read the extended abstract, followed by watching the 12-minute presentation video. The abstract unveiled the scope of the study to students, many of whom did not realize the underlying complexities. Nearly all students responded with statements about gaining clarity about the purpose of the experiment and its role in testing the differences between 2D and 3D graphs. Students commented on how they now understood the purpose of comparing 2D and 3D graphs, and also the potential benefits that may stem from research, such as graphical accessibility to the visually impaired. A bigram plot of the student responses to the abstract reflection prompt is shown in Figure 3.

Lastly, more than half of the students (78.5%) responded that they preferred the presentation over the extended abstract when asked “If you had to hear about this study using only the extended abstract or only the presentation, which one would you prefer? Which one would be better for determining whether the experiment was well designed?” One student responded “I am a visual learner so I would have rather heard about in through the presentation. It also broke down the steps which is easier for me to understand. I think the presentation as a whole would be better for determining how the experiment is designed.” Another student said “Personally I like the abstract better. If I get confused on something it is so much easier to go back and reread to understand what is going on. If I ask myself questions about it, it is much easier to go back and find answers to the questions as well.”

3.4 Discussion

Taken together, our results support the idea that we achieved our goal of providing students of an introductory statistics course with the opportunity to reflect on active research. Students generally appreciated the progressively revealing nature of the graphics project, which is evident from the abstract and presentation reflections. When provided with the post-experiment reflections, students often either missed the research objectives of the experiment or had partially correct responses. However, this was expected since the design of the experiment exceeds the scope of experimental design taught in typical introductory statistics courses.

The abstract reflection received many responses indicating that students had moments of realization about the true nature of our research goals, which was further expanded with the presentation reflection prompts. Across all reflections, students were thoughtful, and sometimes amusing, with their responses and they were on the path of statistical thinking.

Having students participate in experiments is not uncommon, possibly attributed to the readily available convenient samples within academia and potential applications to introduce new course material (Margaret, 1994). However, these studies typically end for students after completing the experiment (Fisher et al., 1997a; McGowan, 2011; Zacks et al., 1998). In our study, we integrated the graphics experiment as a course project with the addition of reviewing research material.

What we found was that the graphics project found success within the recommendations of the GAISE College Report by using real data to illustrate the contextual purpose of scientific research (Carver et al., 2016). Students were able to demonstrate their ability to think statistically through the series of reflections. The graphics project made use of real data, along with data collection, within the scope of an approachable and field-related topic, which allowed students to see how scientific research is conducted in the field of statistics.

A limitation of our study is the use of open-ended responses that do not objectively assess student learning. While the student responses were useful in gathering insight, the responses are widely varied and do not have direct comparisons of statistical thinking throughout the modules. Another limiting factor is the tiered layering of convenience sampling, with instructors being recruited before recruiting students, which impacts the generalization of our findings. Nonetheless, the 82 students who participated provided meaningful answers that displayed some level of statistical thinking throughout the graphics project.

Future studies could use a similar framework to conduct experiments on more typical 3-dimensional data, such as heatmaps. The use of graphical experiments in the classroom not only provides a readily available convenience sample but also adheres to the recommendations of the GAISE College Report (Carver et al., 2016). With the framework we provided in this paper, we aim to make adjustments to further improve the graphics experiment and corresponding project as an experiential learning opportunity.

3.5 Figure legends

Figure 1: Visual display of the experimental design for students who participated in the 3D bar charts experiment. Kits of graphs were created by first choosing five ratios from nine available options (1). Each ratio then uses all graph types, with the exception of the 3D-printed graphs for online students (2). Finally, all graphs were randomly assigned to have the marked bars as adjacent or separated (3).

Figure 2: Bigram of student responses (n=82) to the pre-experiment prompt. Each line represents pairs of words that appeared together where each pair occurred at least twice. Students generally understood that science is about investigating research questions and collecting data.

Figure 3: Bigram of student responses (n=63) to the abstract reflection prompt. Each line represents pairs of words that appeared together where each pair occurred at least twice. Students generally focused on the research objective of comparing 2D and 3D graphs.

3.6 Tables

Table 3.1: Questions provided to students in each project module.

Reflection	Question	Prompt
Pre-Experiment	Q3	In this class, you'll be learning about the process of scientific investigation. What do you think that process looks like, from the perspective of a researcher, compared to what it looks like from the perspective of someone in the general public who is a consumer of scientific results? Write a paragraph (at least 3-5 sentences) about how you think science happens.
Post-Experiment	Q5	What do you think the purpose of the experiment was?
	Q6	What hypotheses might the experimenter have been testing?
	Q7	What sources of error are involved in this experiment?
	Q8	What variables were examined? For each variable, identify whether it was quantitative or categorical.
	Q9	What elements of experimental design, such as randomization or the use of a control group, do you think were present in the experiment? Why?
Abstract	Q10	What components of the experiment are clearer now than they were as a participant? What questions do you still have for the experimenter? Write 3-5 sentences reflecting on the abstract.
	Q11	How did the information you gained from the components of this project (participation, post-study reflection, extended abstract, presentation) differ?

Presentation	Q12	What components were emphasized in the presentation that weren't emphasized in the abstract? Why do you think that is?
	Q13	What critiques do you have of this study and its design? What would have made the study better?
	Q14	If you had to hear about this study using only the extended abstract or only the presentation, which one would you prefer? Which one would be better for determining whether the experiment was well designed?

Table 3.2: Number of valid student participants by semester.

Semester	Number of Sections	Number of Students
Summer 2023 (May-June)	1	17
Summer 2023 (July-Aug)	1	23
Fall 2023 (May-June)	1	42

Students under 19 years of age or did not consent were exluded from data collection. To comply with IRB, no demographic information was collected to keep students anonymous with their reflections.

Chapter 4

Graphical Perception of Heatmaps: Dimensionality, Medium, and Accuracy of Ratio Estimations

4.1 Introduction

In the 20th century, advancements in technology made data visualizations increasingly more affordable and accessible to a broader population. The primary change in formal data visualizations was from hand-drawn charts to computer-rendered charts (Tukey, 1965), and with subsequent advances in color printing and available software have allowed for data visualizations to enter other mediums. These include the ability create virtual and physical charts that effectively use the 3-dimensional (3D) world around us with the novel use of 3D-printed charts. This newer type of visualization has gained a small amount of traction in recent years as a method of producing tangible charts (Hu, 2015; Huron et al., 2023; Stusak et al., 2016).

In many disciplines, 3D-printed visualizations have shown mixed or promising results compared to alternative rendering formats. Although a comprehensive review of 3D-printing in scientific fields is beyond the scope of this paper, we highlight several examples of its use for visualization. Katsioloudis & Jones (2018) conducted a study with engineering students and found no statistically significant difference between computer-rendered and 3D-printed models when students were asked to draw a cross-section of a dodecahedron, suggesting similar levels of spatial understanding between digital and physical representations. In contrast, other applications have highlighted advantages of 3D-printed models. For example, Holloway et al. (2019) reported positive feedback from visually impaired individuals when 3D-printed maps were used to support navigation. Similarly, in clinical education, Muff et al. (2022) found that 3D-printed anatomical structures were well accepted alongside VR glasses and 3D computer renderings. As 3D-printed visualizations become increasingly common across scientific fields, we explore their potential use in communicating statistical graphics.

At this time, there are few studies that evaluate 3D-printed charts for the purpose of displaying statistical information (Hu, 2015; Jansen et al., 2013b; Stusak et al., 2015b; Stusak et al., 2016). This may in part be due to the cost of materials and rendering times as compared to charts produced on a computer. A single chart can take up to a day to print, limiting the ability to quickly produce visualizations. Additionally, there are few options for producing such charts from statistical software. In R, the `rgl` package (Murdoch & Adler, 2023) can export STL files that can be used by 3D-printing software, and similarly for the `rayshader` package (Morgan-Wall, 2024). However, these options are not optimized for essential considerations required of 3D-printed charts, such as embossed labels. Given the nature of 3-dimensional data, viewing a 3D realization of statistical information in a 3D environment is a realistic scenario that could increase the ability for information extraction.

4.1.1 Literature Review

Using an identical dataset across multiple chart types does not ensure that the data is perceived in the same way (Cleveland & McGill, 1984a; Hofmann et al., 2012; Vanderplas et al., 2020a). One of the main factors in this phenomenon is how data is encoded into the chart. These encodings include, but are not limited to, placement along axes, lengths, areas, volumes, and color scales. For example, bar charts and pie charts are two common visualizations that have long been the topic of debate (Cleveland & McGill, 1984a; Croxton & Stryker, 1927a; Eells, 1926b; Lewandowsky & Spence, 1989). Bar charts are constructed with the encodings of length, position, and area. In contrast, pie charts make use of arc lengths, angles, and area. Inherently, the comparison of different chart types is a comparison of encodings due to the changes in how data is being displayed.

Cleveland & McGill (1984a) noted that estimates involving numerical accuracy may decrease when increasing dimensionality of the encoding, although this was not formally tested in their experiments. Their reasoning was due to Stevens' power law, a mathematical formulation of how magnitudes are perceived given different stimuli sources (Stevens & Stevens, 1986). The general form of the law is $\psi(I) = kI^\alpha$, where I is the magnitude of a stimulus, $\psi(I)$ is the perceived magnitude, k is a proportionality constant from the unit of the stimulus, and α is the exponent from the type of stimuli used. Studies have estimated that lengths are perceived without bias (i.e., $\alpha = 1$), but areas and volumes tend to have skewed perceptions (Cleveland & McGill, 1984a). This indicates that lower-dimensional charts may perform better when readers are tasked with extracting numerical estimates from the chart, although this has been a topic of debate in the last several decades.

There are mixed results in regard to the use of 3D charts, mostly attributable to the use of the third dimension. When the extra dimension does not convey meaningful information, estimates of accuracy decrease and solution times increase as compared to equivalent 2D charts (Fischer, 2000; Fisher et al., 1997a; Zacks et al., 1998). The same increase in solution time is seen when the third dimension is utilized for displaying data, but can sometimes produce better error rates than 2D charts (Barfield & Robless, 1989a; Kraus et al., 2020). Additionally, when given the option of 2D or 3D charts for extracting numerical information, the 2D charts showed increased preference and confidence than their 3D counterparts (Barfield & Robless, 1989a; Fisher et al., 1997a). It is worth noting that all of these studies use 2D or virtual reality renderings of 3D charts and not physical 3D charts.

Many studies that assess 3D charts use computer-rendered displays that leverage high levels of precision in their output. Digital renderings allow for exact control over visual properties, whereas physical charts are subject to factors such as chart materials, lighting conditions, interactivity, and viewing distance. Some of these factors have already been suggested to have a measureable effect within computer renderings (Tarr & Kriegman, 2001; Wang et al., 2022). While virtual reality can offer similar interactions with 3D objects compared to the real world, perceptual judgements can be distorted in virtual environments (Park et al., 2021; **kelly2018?**). The result is that many findings for the perception of 3D statistical graphics are limited to the method in which they were rendered.

To date, few empirical studies have investigated 3D-printed charts, and it is unclear if they follow the framework of existing theories on data visualizations. One of the first studies to examine digital and physical renderings primarily focused on information retrieval times, where physical 3D heatmaps had lower completion times than digitized 2D and 3D alternatives (Jansen et al., 2013b). In their study, error rates were also collected, but had small effect sizes due to the 2D, digital 3D, and physical 3D charts. Other studies on physical statistical graphics tend to focus on bar charts, where the third dimension is not used for information. Physical 3D bar charts are suggested to have conditionally improved memorability, subject to specific datasets (Stusak et al., 2015b; Stusak et al., 2016). Hu (2015) discussed the construction of 3D-printed bar charts with accessibility features, though these designs do not appear to have been empirically evaluated.

We hypothesize that 3D charts in 3D environments will produce better information extraction than their computer-rendered counterparts. Specifically, we compare 3D-printed charts to digital renderings of 2D static charts and 3D interactive charts. We suspect that this difference will hold across multiple data sets and different magnitudes of stimuli. In this paper, we evaluate the accuracy of numerical estimations on true 3D charts by conducting a factorial experiment that assessed chart types and ratios of pairs of stimuli. We discuss the construction of the stimuli and how we closely matched the charts to compare 2D, 3D-digital, and 3D-printed renderings of heatmap data.

4.2 Methods

Our study was designed to evaluate and expand the literature on numerical estimation of dimensionality for 3D charts. In our study, we focused on 2D and 3D heatmaps, which we carefully constructed to ensure that differences can be attributed to the dimensionality of the chart and the mapping of data when comparing 2D to 3D. All of our data and methods are publicly available for open science and reproducibility at `https://github.com/TWiedRW/ch3-heat3d`. In this section, we discuss the process of designing our experiment and participant recruitment.

4.2.1 Stimuli

We define stimuli as the magnitude of our selected values. In a 3D Cartesian space, the X- and Y- axes represent the coordinates of the stimuli, and the Z-axis represents the value for the stimuli. Each X and Y coordinate is represented by a square tile with a 1:1 aspect ratio. In a 2D space, the Z variable is mapped to fill with a monochromatic gradient. All stimuli and remaining randomly generated values range between 0 and 100 units. In this experiment, $X = 1, 2, \ldots, 10$ and $Y = 1, 2, \ldots, 10$.

Our experiment is designed using the method of constant stimuli, where comparisons between stimuli are with respect to a stimulus that remains the same magnitude (Kingdom & Prins, 2016, 3). We set the constant stimuli at 50 units. For stimuli between 50 and 100, we set the maximum stimuli value at 90 and equally partitioned the ratios of stimuli with the constant stimuli, $\frac{50}{90} = 0.556$ to $\frac{50}{50} = 1.0$, resulting in 4 varying stimuli values where 50 is the smallest value. The same ratios obtained with stimuli between 50 and 90 were used to create 4 stimuli varying between 0 and 50, where 50 is the largest value. Additionally, we also included a stimulus pair where both values are 50, resulting in nine total pairs of stimuli. All stimuli values can be found in Figure 4.1.

Two datasets were created for the experiment. To generate non-stimuli random values, we used a mixture distribution of random uniform noise and a mathematical function to populate our coordinate grid. The mathematical functions are scaled between 0 and 100, $g(Z) = 100 \cdot \frac{Z - \min(Z)}{\max(Z) - min(Z)}$. The first dataset, set 1, used the formula for the top half of a sphere that is centered within our X and Y coordinate grid, $f_1(X, Y) = \sqrt{7^2 - (X - \bar{X})^2 - (Y - \bar{Y})^2}$, where $\bar{X}$ and $\bar{Y}$ are the averages of the X and Y coordinate ranges. The second dataset is calculated similarly using the

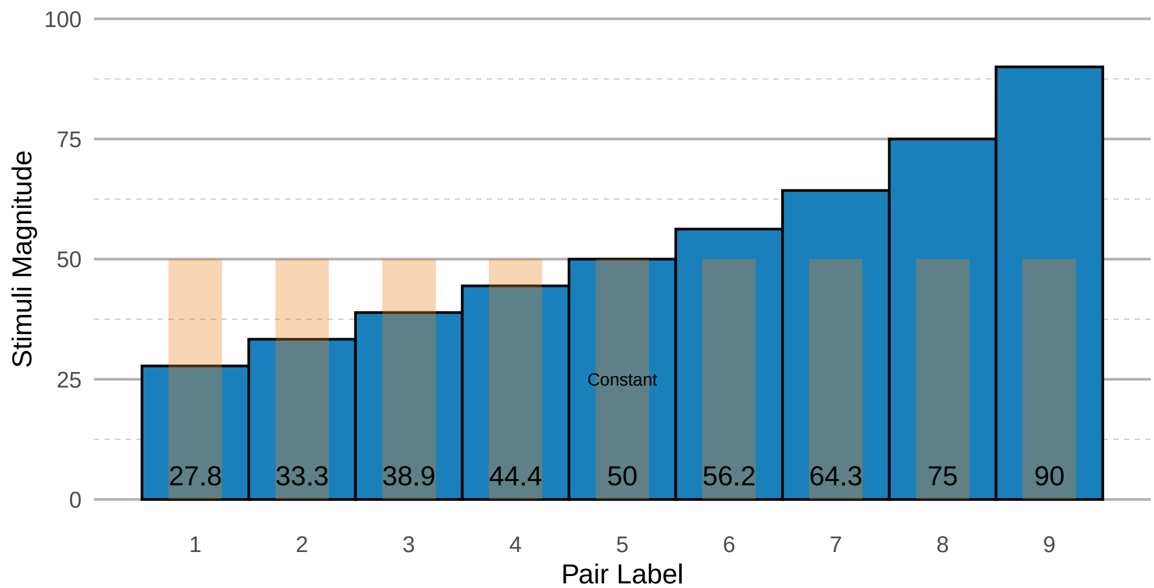
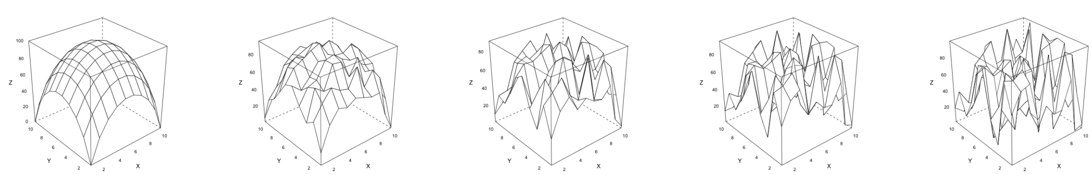


Figure 4.1: Values for stimuli in the heatmap experiment. All values are paired with the constant stimuli of 50 units, creating nine pairs of stimuli.



(a) $c = 0$ (b) $c = 0.25$ (c) $c = 0.5$ (d) $c = 0.75$ (e) $c = 1$

Figure 4.2: Mixture distribution of Equation 4.1 using the formula for the top half of a sphere. As c approaches 1, the distribution resembles uniform random noise.

formula for the bottom half of a sphere, $f_2(X, Y) = \sqrt{7^2 - (X - \bar{X})^2 + (Y - \bar{Y})^2}$. Denoting Z as the random values, $U(0, 100)$ as a random variable drawn from a continuous uniform distribution between 0 and 100, and X, Y as heatmap coordinates, our random heatmap data is calculated in Equation 4.1 with $c = 0.3$. An example of varying c values is presented in Figure 4.2.

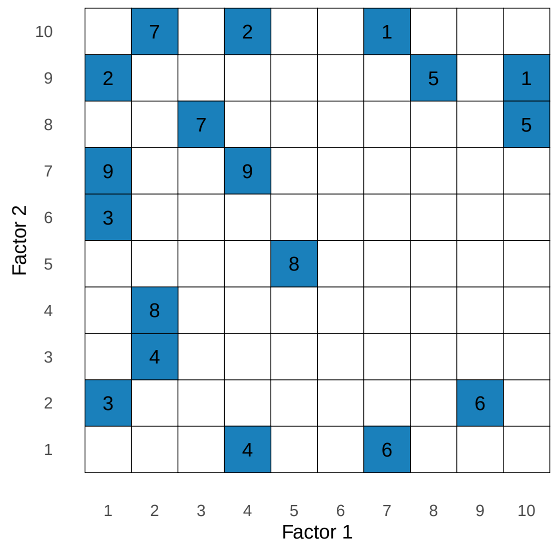
$$Z = c \cdot U(0, 100) + (1 - c) \cdot g(f_i(X, Y)) \tag{4.1}$$

The placement of stimuli values onto the randomly generated heatmap data was done via simulation to try to make their placement look as natural as possible. Twenty random heatmaps were generated. For each heatmap, the non-constant stimuli was placed onto the coordinate such that the difference between the stimuli and randomly generated value is minimized. The constant stimuli was then placed onto a coordinate within a Manhattan distance of three or four that minimizes the difference between the constant stimuli and randomly generated values, where the Manhattan distance is given by $|X_i - X_j| + |Y_i - Y_j|$ for stimuli i and j. To ensure that stimuli placement is evenly positioned across the heatmap, the count of stimuli was computed separately across the X and Y axes. For example, in Data Set 1, the X-axis has four stimuli in $X = 1$, three stimuli in $X = 2$, and so forth. Chi-squared statistics were calculated for each axis and the heatmap with the smallest average Chi-squared statistic was selected as the final dataset. A visual inspection of this process showed that the stimuli were not clustered in any one area of the chart and that the stimuli look natural with respect to the random mixture distribution.

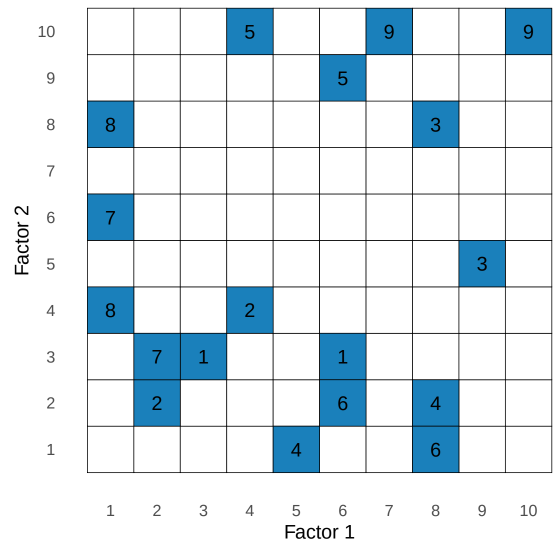
4.2.2 Charts

Three types of charts were considered for this study: digital and static 2D (2dd), digital and interactive 3D (3dd), and 3D-printed (3dp). We constructed these charts so that they were as similar as possible, but inherent differences in dimensionality led to artistic decisions that attempt to focus solely on the dimensionality of the charts. The process of creating the charts is discussed in this section.

The 3D-printed charts were rendered with OpenSCAD (Kintel, 2023). To include plot text, a 120mm by 120mm by 10mm base was created with a solid color that was either white or black. Cells of the heatmap measured 10mm by 10mm,



(a) Placement of stimuli on Set 1

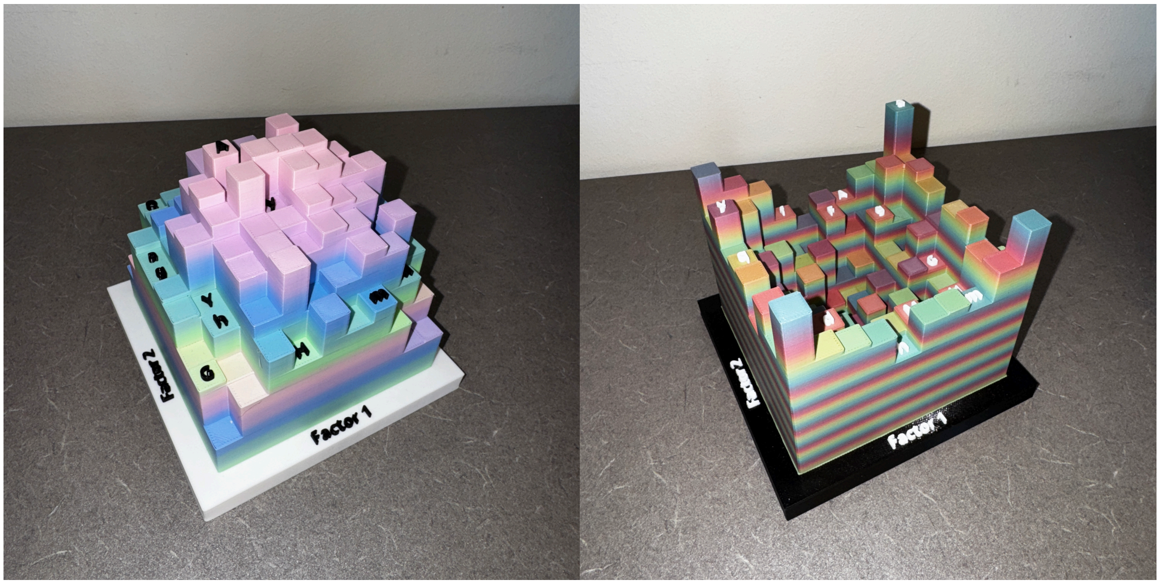


(b) Placement of stimuli on Set 2

Figure 4.3: Placement of stimuli on the two data sets. Tiles with solid fills are the fixed stimuli values, where white tiles are randomly generated.

resulting in a heatmap that is 100mm by 100mm and is centered on the base. The upper bound of the height of heatmap values is 100mm, where 1-unit in the heatmap data is represented by 1mm of height on the heatmap. Once rendered, the heatmap was saved to 3D Manufacturing Format (3mf) and Standard Triangle Language (stl) files. A variety of solid and gradient filaments were tested for the bars of the 3D prints, and we eventually settled on a rainbow gradient. Some charts were printed with a white base and black text, and other charts had a black base with white text. An example of the 3D-printed chart is shown in Figure 4.4.

To closely match the 3D-digital chart to the 3D-printed chart, multiple stl files were created for each colored component and combined with the RGL package (Murdoch & Adler, 2023). The base was rendered with white smoke (#F5F5F5) to slightly contrast with the default white background color (#FFFFFF). Heatmap tiles were rendered with cyan (#74CCFF) and text labels were rendered with black (#000000). Lighting was fixed at 45 degree angles at two opposite corners of the



(a) Data Set 1 (b) Data Set 2

Figure 4.4: 3D printed heatmaps.

chart. The end result was a near-perfect replica of the 3D-printed charts, with the exception of different heatmap tile colors Figure 4.5.

Unlike the 3D charts, the 2D charts needed a different encoding to convey heatmap values. The 2D heatmaps were created with `ggplot2` (Wickham, 2016) using `geom_tile()`. Fill colors for the cells use a color gradient ranging from Blue Zodiac (#0C2841) to Malibu (#66D9FF), which were selected from a color picker using shadows on our initial 3D-printed charts. The color interpolation was performed with the `scale_fill_gradient()` function from the `ggplot2` package.

4.2.3 Subject Recruitment

Participants were recruited from STAT 218: *Introduction to Statistics* at the University of Nebraska-Lincoln in the summer and fall of 2025. The experiment was incorporated into the curriculum as a project that allowed students to get hands-on experience with statistics. For data to be collected as part of our study, students

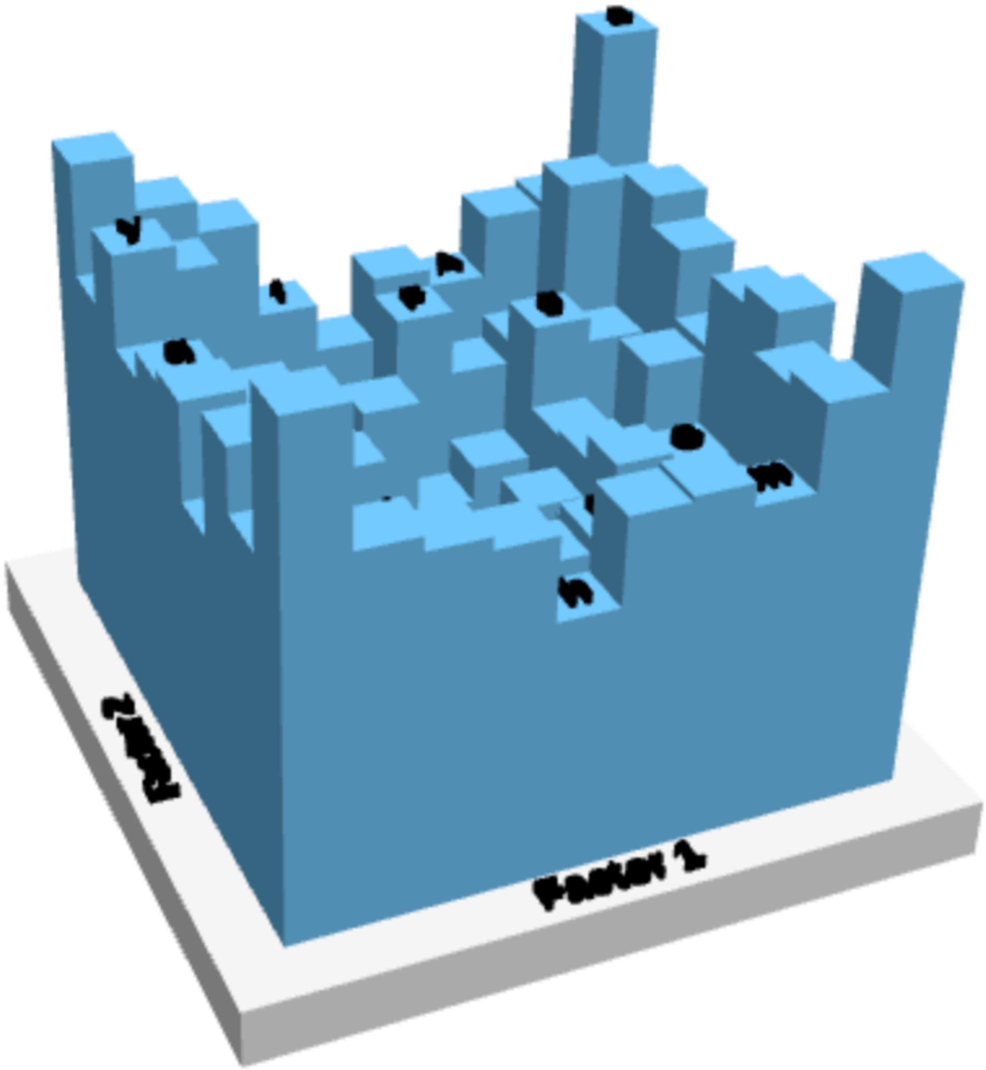


Figure 4.5: RGL rendering of Data Set 2 for 3dd charts.

Figure 4.6: Color palette for 2D-digital charts. The colors are interpolated from #0C2841 to #66D9FF, which are the colors of the lighting conditions for a 3D-printed chart created with cyan filament.

had to be at least 19 years old (age of majority in Nebraska) and consent to participation.

STAT 218 is a general education statistics course at the University of Nebraska. As such, many students enrolled in the course are in the 18-25 age range and are in the process of obtaining a Bachelor's degree. This allowed for the unique opportunity for us to obtain a sample size that is larger than typical studies in graphical perception, and that could accommodate in-person participation required for the 3D printed charts. Some sections of STAT 218 were held online in an asynchronous format.

4.2.4 Experimental Design

Our experiment was designed with a 3 media type x 2 datasest x 9 comparison treatment structure. Media type is our main interest, with 2D-digital, 3D-digital, and 3D-printed charts. To ensure that results are not confounded with datasets, we used two datasets, with a total of 9 stimuli pairs in each dataset. The order of treatments was given so that media and dataset combinations were grouped together randomly in the sequence with stimuli pairs randomized within the groupings. This meant participants did not have to switch charts as frequently, reducing the risk of using the wrong chart. For students enrolled in online sections of STAT 218, the 3D-printed charts were removed from their trials as they were not required to be on campus.

Due to practical constraints, stimuli pairs were incompletely blocked. A full factorial design would result in 54 trials per participant, which could lead to a decrease in quality responses (Herzog & Bachman, 1981). Therefore, we selected four out of the nine possible stimuli pairs to create incomplete blocks for each chart. This resulted in 18 blocks for a balanced incomplete block design. Within each block, media type is fully crossed with dataset. Using the incomplete block structure, participants completed a total of 24 trials, which is more practical than the full factorial design.

Following the procedure of Cleveland & McGill (1984a), we asked participants two questions in each trial. The first question asked participants which stimuli in a specified pair is larger or if the stimuli are the same value. Next, participants were asked to estimate the magnitude of the smaller stimuli if the larger stimuli represents 100 units, which is a subtractive process (Veit, 1978). This question is designed to estimate A/B, where $A \leq B$.

4.2.4.1 Shiny Application

A Shiny application (Chang et al., 2023) was developed to administer the experiment. The application consisted of five sections: informed consent, demographics, practice, experiment, and wrap-up. The entire application was designed to be completed in approximately 30 minutes.

The Shiny application started with the informed consent screen, allowing participants to select if they are a STAT 218 Student and if they agree to the data collection. Participants had to select a data collection option to continue. After submitting their data collection response, a completion code was generated and saved on the last page of the application. This code was required for class, but participant consent was only required to save the data, so underage participants could still get the code for course credit.

If participants agreed to have their data collected, they were presented with the demographics section. This section asked participants about their age, gender identity, highest education level, and how their participation in the experiment is graded. The last question was a text box and asked participants to specify their favorite movie and/or actor. Demographic information was combined with the application start time and completion code to create a hash for an anonymized participant identifier using the `rlang` package (Henry & Wickham, 2025).

Once a participant completed the demographics page or selected “No” to the data collection question, they were given a practice page. A modal dialog window was initially shown with instructions, and users could display this window again at any point during the practice trials. The practice consisted of four questions: two trials from 2dd charts and two trials from 3dd charts. Once participants provided a response to both questions in a practice trial, they were able to display the correct solution to check their answers. Since the practice trials were only to inform participants about the mechanics for answering the questions, we did not include 3D-printed charts. This eliminated the possibility of participants using a practice chart for the experiment.

The experiment page was presented to participants after completion of the practice trials Figure 4.7. Each page contained two questions – one question for identifying which stimuli is larger and another question for estimating the value of the smaller stimuli if the larger stimuli is 100 units. Each trial had the initial slider position randomized in order to prevent starting position bias. For 3dd charts, the number of interactive clicks was also recorded. A trial could only be submitted if the first question is answered and if the slider was moved at least once.

After completing all trials, participants were given the completion code and informed to provide the code to their instructor since they would not have access to the code after closing out of the experiment.

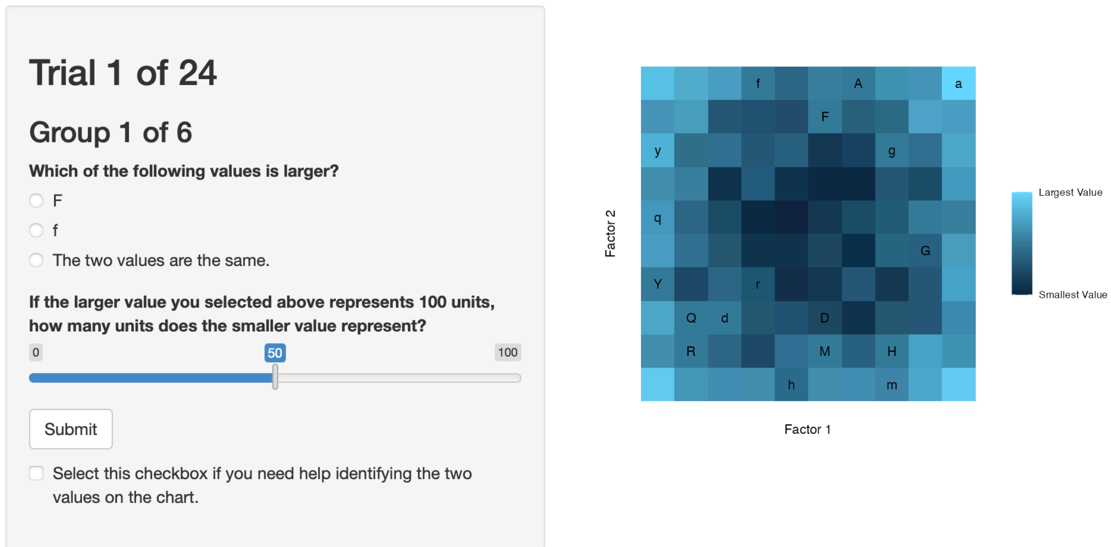


Figure 4.7: Graphics experiment page. Each trial contained two questions about a pair of stimuli, and a check box that displays the location of the stimuli if needed.

4.2.4.2 Method of Analysis

The design of our experiment is a balanced incomplete split-plot with replication, where media type and dataset are whole-plot factors for each participant and stimuli pair is the split plot. A similar design was presented by Mandal et al. (2020) for a single replicate. For our analysis, we used the following model as a baseline:

$$Y_{ijklm} = \mu + S_i \times M_j \times P_k + \gamma_{lm} + \omega_{ijlm} + \epsilon_{ijklm}$$

In our model, Y_{ijklm} is a metric of error for participant responses to the ratio estimation question, S_i is the effect of dataset i, M_j is the effect of chart type j, P_k is the effect of stimuli pair k, and $S_i \times M_j \times P_k$ includes all corresponding main effects and two-way and three-way interactions. Our random effects include $\gamma_{lm} \sim N(0, \sigma_A^2)$ for participant m in block l, $\omega_{ijlm} \sim N(0, \sigma_B^2)$ for the ordering of dataset i and chart type j for participant m in block l, and $\epsilon_{ijklm} \sim N(0, \sigma_C^2)$ for the residual error. We assume that γ_{lm}, ω_{ijlm}, and ϵ_{ijklm} are independently and identically distributed.

The first question in each trial was to indicate which value within a stimulus pair was larger, or if they were the same value. Following Cleveland & McGill (1984a), we use this question as an attention check to verify that participants are following the procedure and that the correct comparison is being made.

4.3 Results

In this section, we describe the demographic information of our sample, cleaning of participant responses, analysis of responses, and interactions participants had with the Shiny application.

4.3.1 Demographics

Our experiment was conducted with eight sections of Stat 218 between August 25th and December 19th in 2025 at the University of Nebraska. To participate in the experiment, students were instructed to visit communal office hour locations to access the 3D printed charts. We recorded 482 interactions with the shiny application, of which 259 students completed the entirety of the experiment. Of these students, 92 were enrolled in online sections of the course and did not have access to the 3D printed charts. The vast majority of participants were actively completing their undergraduate degree and between the ages of 19 and 25. Full demographic information can be found in Table 4.1.

Almost all participants completed the experiment in under 20 minutes. Summary statistics can be found in Table 4.2, with the overall distribution of times shown in Figure 4.8. Individual trials had a median completion time of 0.23 minutes.

Table 4.1: Demographic Information

	Count	Proportion
Gender		
Female	166	0.64
Male	90	0.35
Variant/Nonconforming	1	0.00
Prefer not to answer	2	0.01
Age		
19-25	252	0.97
26-30	2	0.01
31-35	0	0.00
36-40	1	0.00
41-45	3	0.01
Prefer not to answer	1	0.00
Education		
High School or Less	23	0.09
Some Undergraduate Courses	214	0.83
Undergraduate Degree	12	0.05
Some Graduate Courses	4	0.02
Graduate Degree	2	0.01
Prefer not to answer	4	0.02

Table 4.2: Summary of experiment completion times in minutes.

Section	Mean	SD	Median
In-person	8.08	3.39	7.45
Online	5.70	3.42	5.09
All Participants	7.23	3.58	6.86

During our initial exploration of the results, we discovered a non-negligible number of students who did not follow the instructions, which is not surprising in an intro statistics class activity. In some cases, participants appeared to be using alternative estimation strategies, such as estimating the difference between the two values in a stimulus pair, rather than the specified ratio strategy. Additionally, we noticed that several students were randomly entering values in order to obtain the completion code at the end of the experiment. We provide some examples of individual participant responses in Figure 4.9 to illustrate this issue.

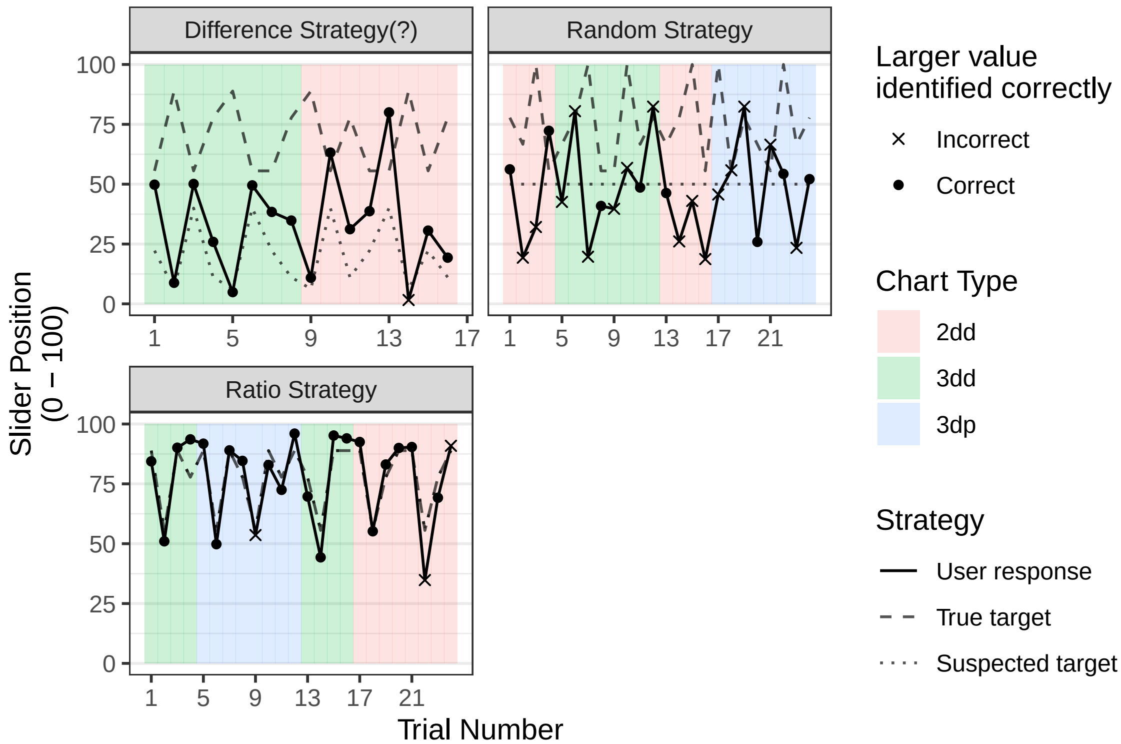
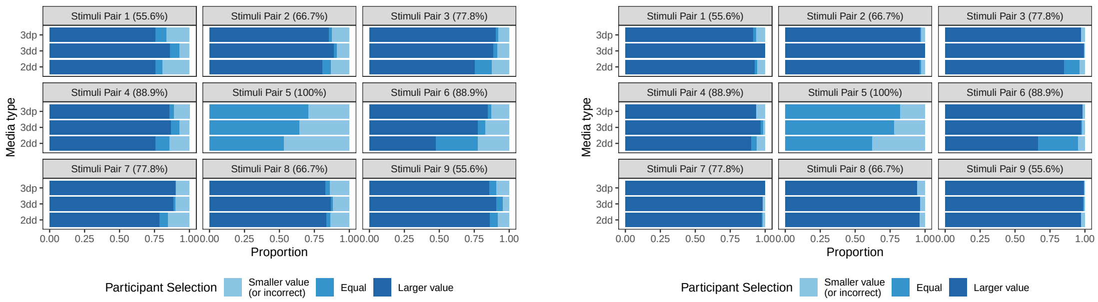


Figure 4.9: Three potential estimation strategies from participants. Many participants followed instructions to estimate ratios. However, some participants appeared to estimate the difference between stimuli pairs or submitted random values.

It is commonly acknowledged in psychology that careless respondents should be addressed to some degree, although there is no general consensus on how to handle this without cherry picking (Jaso et al., 2021; Ward & Meade, 2023). Some proposed solutions include removing participants based on observed patterns or fitting mixture models to address latent class variables. In this section, we discuss one possible subset of participant responses to address careless respondents. We also include several alternative models in the appendix.

4.3.2 Addressing Random Categorical Responses

To improve the quality of our data, we started by assessing random guesses in identifying the larger value of a stimulus pair. In each trial, participants have a 1



(a) No participant filtering

(b) Exclude random guessers

Figure 4.10: Proportion of correct responses to Question 1 across stimuli pairs and media types. For all pairs, with the exception of Pair 5, a correct answer was 'Larger value'. Stimuli Pair 5 had two identical values, meaning that options other than 'Both values are the same' were incorrect.

out of 3 chance for correctly identifying the larger value if they are randomly guessing. Since stimuli pair 5 contains identical values, we temporarily exclude those trials to focus on stimuli pairs where there was a difference between the values. For each participant, we test $H_0 : \pi \leq 2/3$ vs. $H_1 : \pi > 2/3$ with $\alpha = 0.05$, where π is the probability of a participant correctly answering the initial question and α is the level of the test. That is, participants who are following instructions should select the correct option at least twice as often as either incorrect alternative. Given the lack of standardized criteria for identifying careless responding, the one-sided Binomial test should extract users who are suspected to be following the instructions. Using the first question for participant screening, 98 participants are excluded for suspected random selecting, leaving 161 participants. The effect of this exclusion can be seen in Figure 4.10.

4.3.3 Analysis of Error

We follow the same calculation of error as Cleveland & McGill (1984a), which is shown in Equation 4.2. Error is defined as the base-2 logarithm of the absolute estimation error meaning that differences of one unit on this scale correspond to approximately doubling the size of the error. The addition of 1/8 prevents distortion for responses with near-zero error.

$$y = \log_2(|\text{User Response} - 100 \times (\text{True ratio})| + 1/8) \tag{4.2}$$

In our analysis, we excluded individual responses where participants were incorrect in their response to the initial question and all responses for stimuli pair 5. The initial question served as an attention check to verify that the participants were making the correct comparison between stimuli values. For stimuli pair 5, we removed these trials since correct solutions should be a binary response. However, we discovered a discrepancy in how participants were responding to both questions for these trials, which we discuss in the appendix.

With our specified exclusion criteria, we find that there was marginal evidence for the effects of chart type (p-value = 0.041) and the data set and stimuli pair interaction (p-value = 0.073). This finding is consistent with models fitted with various participant and response filters.

Our findings suggest that the 2D heatmaps had evidence of higher log absolute error than either type of the 3D heatmaps. On the base-2 log scale, the error for 2D heatmaps was larger than the error for digital 3D heatmap by 0.414 (95% CI: 0.253, 0.574). This finding is similar for the 3D printed heatmaps, where the log absolute error for 2D heatmaps was larger by 0.413 (95% CI: 0.234, 0.591). There were no detectable differences in error between the digital and physical 3D heatmaps (p-value = 1).

When examining the interaction between stimuli pairs and underlying datasets, 11 differences in log absolute error were detected at the $\alpha = 0.05$ level. Seven of these differences occurred with stimuli pair 6 (ratio = 0.889) and dataset 1, which

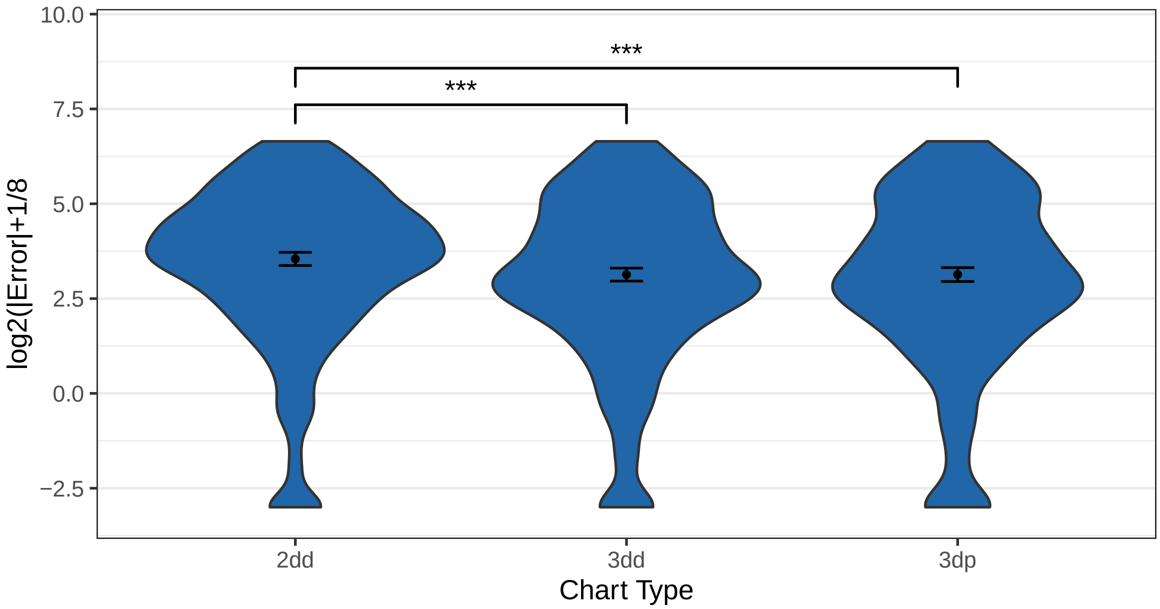


Figure 4.11: Distributions of log error overlayed with estimated marginal means and their 95% confidence intervals. Differences were only detected between 2D heatmap and both types of 3D heatmaps, where the 2D heatmap had evidence of larger errors.

had the smallest estimated marginal mean of log absolute error. All differences can be found in Figure 4.12, where significant differences are assessed via a compact letter display from the `multcomp` package (Hothorn et al., 2008).

4.3.4 Participant Interactions with Shiny Application

In this section, we describe participant interactions with the Shiny application using the same participant exclusion criteria as our model. As participants completed the experiment, we recorded the amount of time to complete each trial, the number of cursor clicks on the digital 3D heatmaps, and the number of times that the slider was moved. We also recorded the initial starting position of the slider, which was randomly placed for each trial.

Many participants completed each trial in under 20 seconds. The median completion time was 14.8372396 seconds, with a standard deviation of 22.9971286

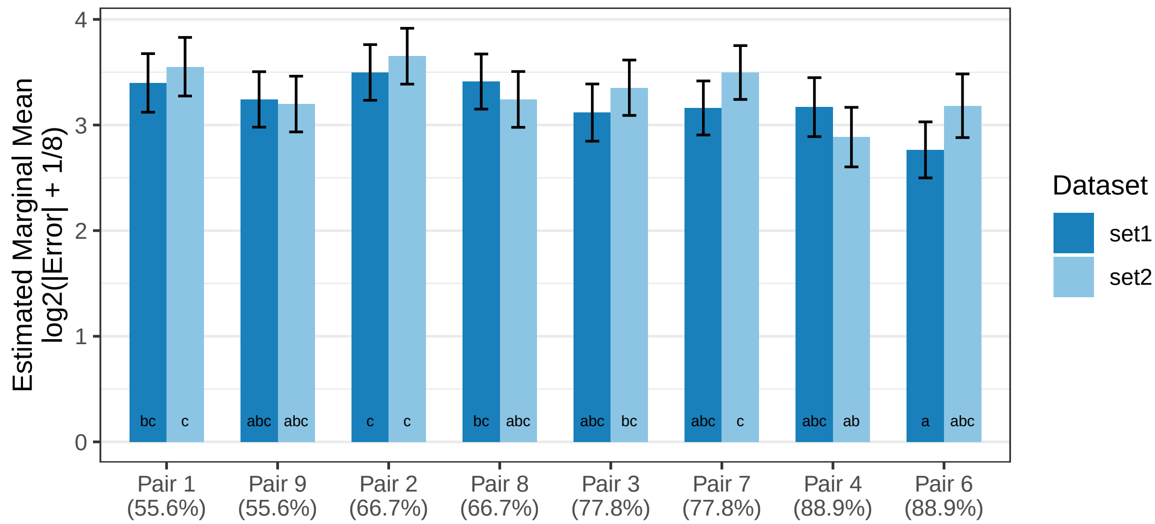
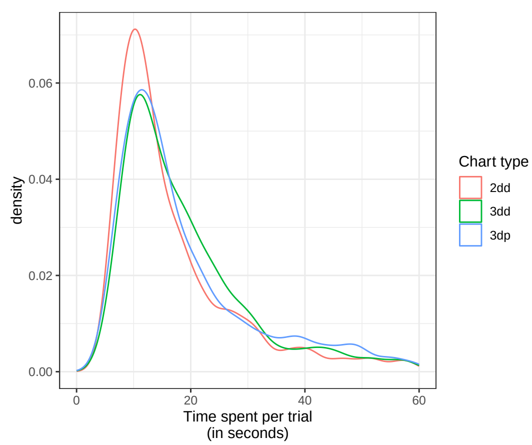
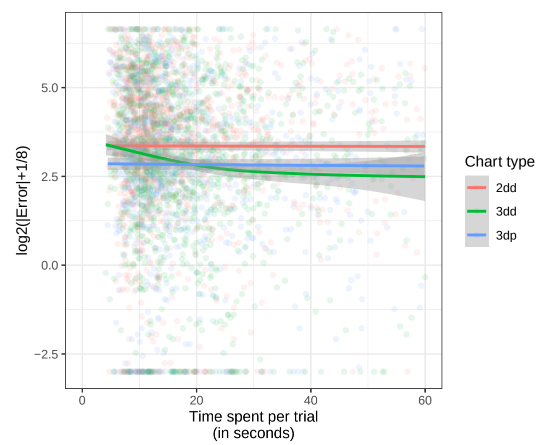


Figure 4.12: Estimated marginal means of stimuli pair and dataset. Bars sharing the same letter are not significantly different from one another.



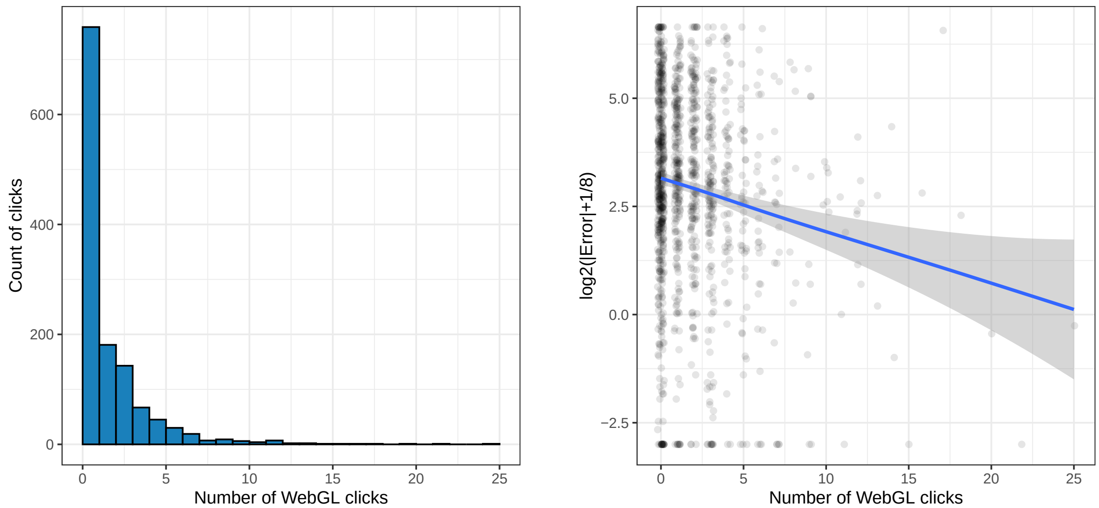
(a) Frequency plot



(b) Time vs. Error

Figure 4.13: Visual summary of time spent on each trial. 137 trials are omitted due to lasting longer than 60 seconds.

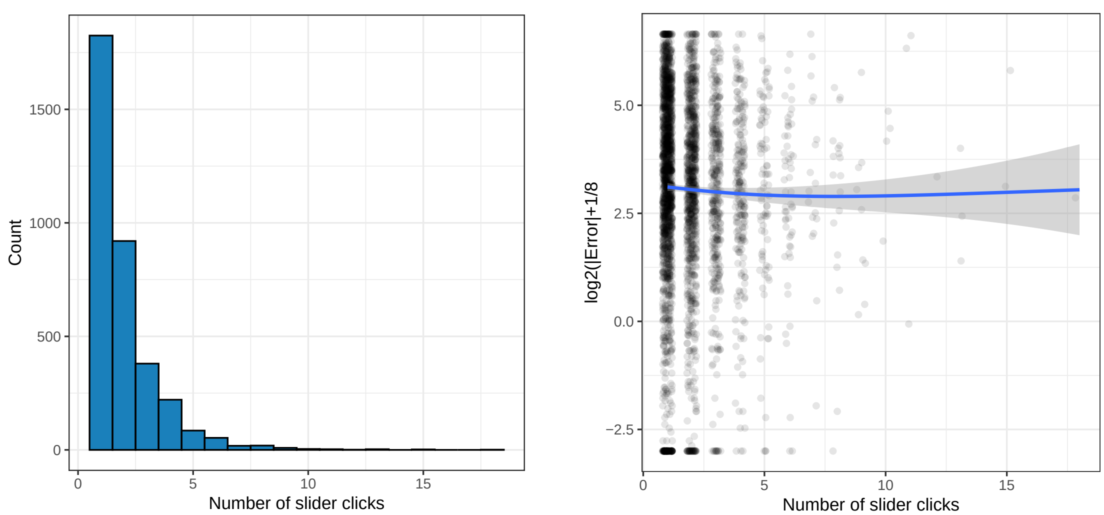
seconds. There were 91 participants who had at least one trial exceed 60 seconds. Due to the design and administration of the experiment, we are unable to account for the cause of longer trial completion times, but it is possible the students may have been interrupted or distracted before returning to the task.



(a) Histogram

(b) WebGL clicks vs. Error

Figure 4.14: Participant interactions with the digital 3D heatmaps.



(a) Histogram

(b) Slider clicks vs. Error

Figure 4.15: Participant interactions with the slider.

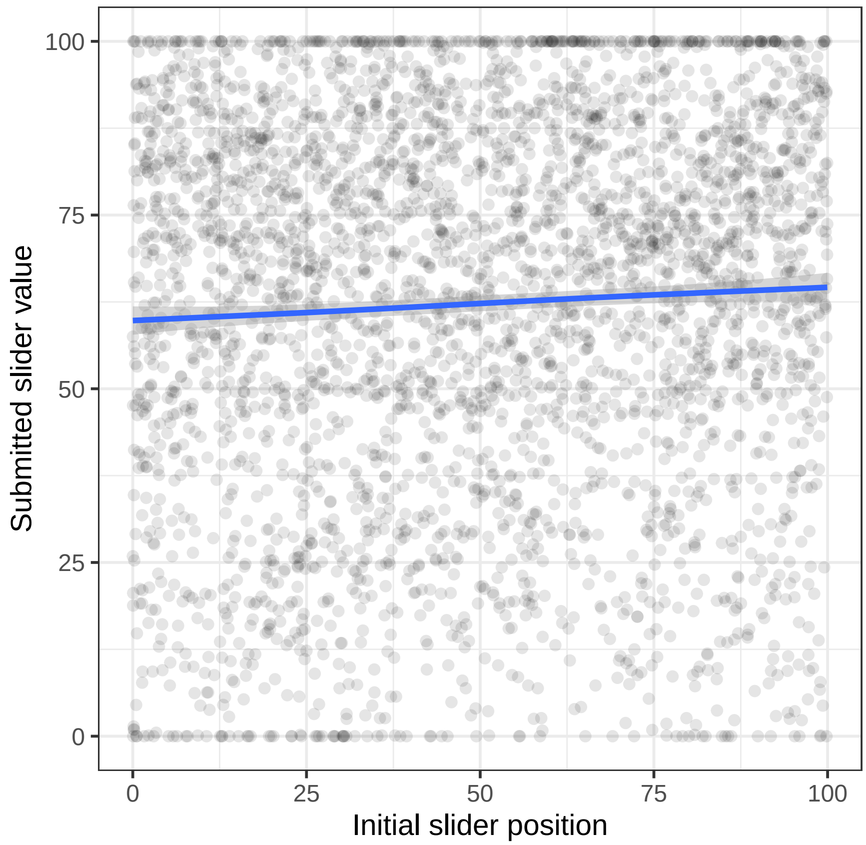


Figure 4.16: Effect of initial slider position on the submitted slider value.

4.4 Discussion and Conclusion

Our study was designed to test the numerical estimation of ratios in 2D and 3D heatmaps rendered in digital and physical environments. The current state of tangible statistical graphics is limited mainly to tactile displays, whereas 3D printed statistical graphics are largely unexplored. We found that when the third dimension is conveying meaningful information, the heights of 3D heatmaps had smaller errors than the color gradient of 2D heatmaps. No differences were detected between the digital and physical 3D heatmaps, which is perhaps attributed to the exact one-to-one design relationship between the digital display and the 3D printed chart. These findings were consistent across multiple levels of fitted models and participant filtering, which reaffirms our belief that 3D charts have value when meaningfully using all three dimensions.

An interesting observation from the stimuli pairs and dataset interaction was from the non-detectable differences. We note that no differences were detected across datasets for a given stimuli pair. While not explicitly tested in our experiment, there have been previous studies that have investigated and found the behavior of neighboring values to have an effect (VanderPlas & Hofmann, 2015; Zacks et al., 1998). This effect would be worth exploring in future studies with 3D heatmaps. Another concern we had when designing our experiment was the lack of direct translation from color scales to heights. The differences between stimuli pairs of identical ratios were non-detectable, which alleviates this concern for future studies. However, this effect would likely be important to consider if assessing just noticeable differences (Lu et al., 2022; Zhang et al., 2023b).

While our hypothesis that 3D-printed heatmaps would outperform digital 2D and 3D heatmaps in numerical estimations of ratios was not supported, we were able to show that 3D charts may have a place in the analysis of data. When the third dimension is used to display information, the mapping of that dimension to height provides better ratio estimations than a color gradient scale. This result has been observed previously in digital 3D heatmaps (Barfield & Robless, 1989a; Kraus et al., 2020), and our findings suggests that this may hold for physical 3D heatmaps.

Chapter 5

In this section, we include multiple models for the transparency of model selection. Visual inspections of participant responses indicate that several participants clearly did not follow instructions by either using alternate estimation strategies or by carelessly responding Figure 4.10. While we should take these cases into consideration, there is no singular solution to suggest a “best” model (Jaso et al., 2021; Ward & Meade, 2023). Our purpose for including these alternate models is to examine model sensitivity to results through various levels of participant and response filtering.

For each model, we exclude stimuli pair 5. In theory, participants who answer this question correctly will select that both values are the same and move the slider to 100. From the included participants, as described in Section 4.3.2, virtually all participants who put the slider at 100 answered the first question correctly. However, nearly 60 percent of responses indicated that the values were the same in the first question but did not move the slider to 100. Several factors could have attributed to this, such as slider sensitivity or participants using alternate estimation strategies. Rather than correcting what we think participants should have answered, we removed this stimuli pair and motivate future research into “just noticeable differences” for 3D heatmaps.

There are multiple levels of participant and response filtering that we

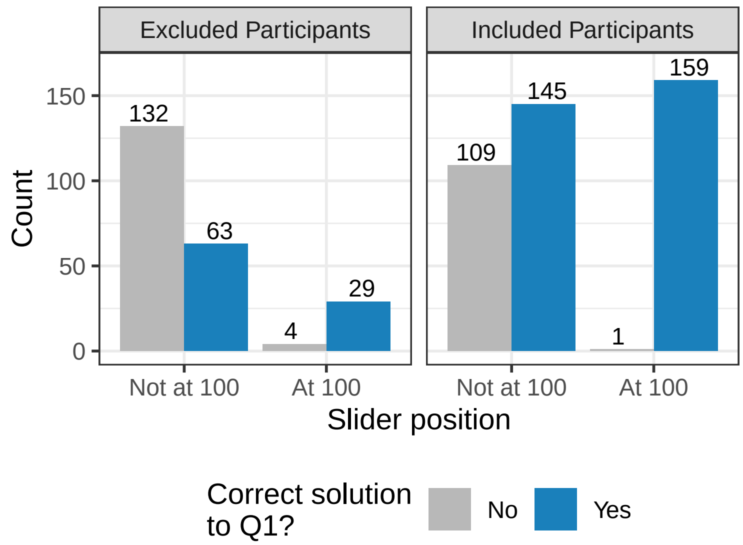


Figure 5.1: Counts of response behavior for Stimuli Pair 5. This pair had identical values, which means that true solutions indicates marking that they were the same value and positioning the slider at 100.

considered. For participant-level filtering, we consider the following levels: (1) no participant filtering and (2) remove suspected random guessers. The process of (2) was described in Section 4.3.2. Filtering individual responses is a straightforward process, where we considered (i) include all responses, and (ii) remove responses with incorrect Q1 solutions.

Table 5.1: ANOVA Table p-values for model terms

Term	All participants, all responses	All participants, Q1 correct	Filtered participants, all responses	Filtered participants, Q1 correct
set	0.340	0.705	0.939	0.735
media	0.010	0.018	0.035	0.041
pair_id	0.244	0.333	0.089	0.063
set:media	0.297	0.711	0.543	0.669
set:pair_id	0.003	0.145	0.001	0.073

Term	All participants, all responses	All participants, Q1 correct	Filtered participants, all responses	Filtered participants, Q1 correct
media:pair_id	0.815	0.740	0.982	0.968
set:media:pair_id	0.489	0.764	0.255	0.833

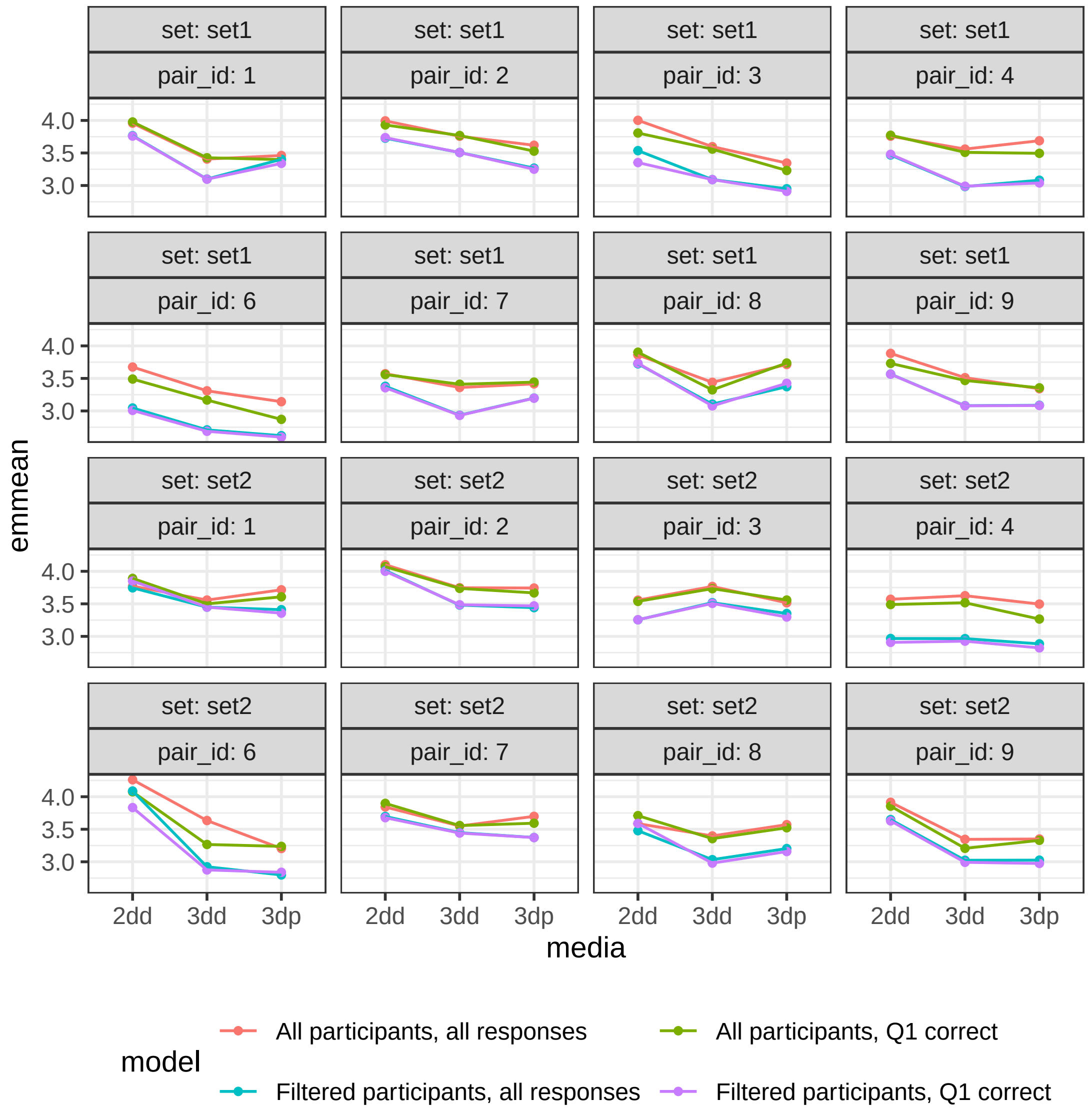


Figure 5.2: Interaction plots for media type and response filtering, facetted by stimuli pair and dataset.

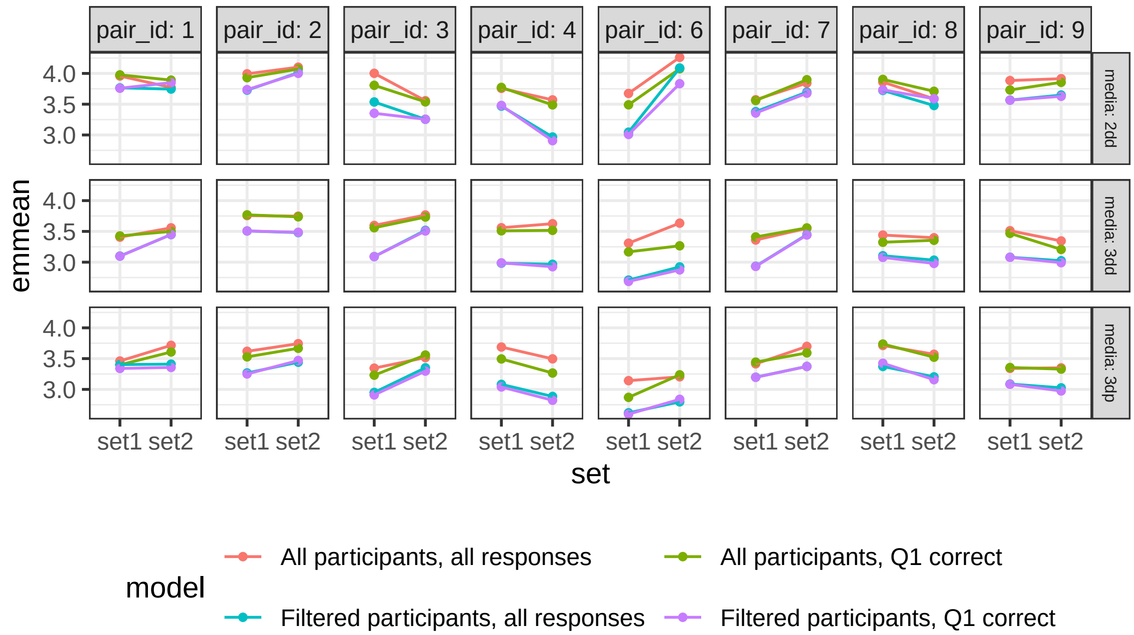


Figure 5.3: Interaction plots for dataset and response filtering, facetted by stimuli pair and media type.

Note

This paragraph is pushed to a new page since the ANOVA table p-values is causing formatting issues

Visualizations of the simple effects for media types, datasets, and stimuli pairs (Figure 5.2, Figure 5.3, Figure 5.4) shows that the difference between participant filtering is mostly a linear shift in log error. Removing participants deemed to be random guessers tended to have smaller errors than including all participants. The shift is less prevalent across participant filters when filtering out incorrect responses to the first question in each trial. This, in addition to the p-values from **?@tbl-all-models-anova**, suggests that our sample size was large enough to counteract effects from careless respondents.

We also attempted to fit Gaussian mixture models to the submitted responses using the `flexmix` package (Grün & Leisch, 2007) to account for careless

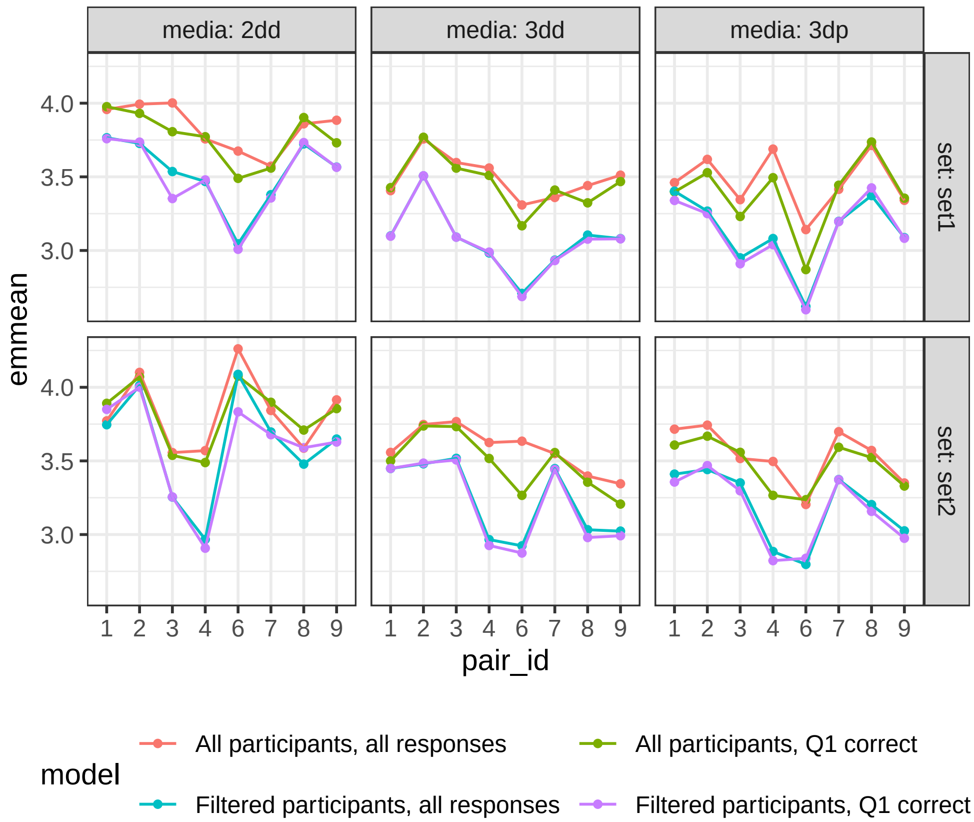


Figure 5.4: Interaction plots for stimuli pairs and response filtering, facetted by dataset and media type.

respondents (Jaso et al., 2021). However, these models only reliably detected a small subset of participants who were would be considered high performers. That is, participants who had a near 1:1 relationship with the ratio estimations strategy and small variation. Visual inspections of the mixture model components did not provide reliable classifications of estimation strategies, so we opted to exclude mixture models from consideration.

5.0.1 Generalized Additive Model

In addition to our parametric methods, we also present the results of a generalized additive model (Hastie, 2017). Here, we define the model similarly to the parametric models, but replace stimuli pair identifiers with their respective numerical ratio value. Thin-plate regression splines in the generalized additive model account for nonlinearity which could alleviate issues due to different estimation strategies. The formal statistical model is as follows:

$$y = \mu + S_i \times M_j + s_i^S(R) + s_j^M(R) + P_m + \epsilon \tag{5.1}$$

where

- $y = \log_2(|\text{Error}| + 1/8)$
- $S_i \times M_j$ is the fixed effects for main effect and two-way interactions of dataset S_i and media type M_j
- $S_i^S(R)$ is a thin-plate spline function for ratio accounting for dataset
- $S_j^M(R)$ is a thin-plate spline function for ratio accounting for media type
- P_m is the random effect of the m^{th} participant
- ϵ is random error

A generalized additive model was fit using the `mgcv` package (Wood, 2017b) on the entire dataset and the filtered dataset. The data for these models remove stimuli pair 5 and incorrect responses to the initial question. In both cases, results of the generalized additive models suggest that the 2D heatmaps have larger errors than the 3D heatmaps, with no differences detected between the digital and physical heatmaps **?@tbl-gam-emmeans?@tbl-gam-diffs**.

Table 5.2: Parametric coefficients in gam model with all participants.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	3.773	0.083	45.613	0.000
setset2	0.014	0.080	0.172	0.863
media3dd	-0.317	0.077	-4.094	0.000
media3dp	-0.389	0.089	-4.359	0.000
setset2:media3dd	0.018	0.110	0.163	0.871
setset2:media3dp	0.078	0.124	0.630	0.529

Table 5.3: Approximate significance of smooth terms in gam mode with all participants.

Smooth	edf	Ref.df	F	p-value
s(target_ratio):media2dd	1.002	1.003	0.058	0.813
s(target_ratio):media3dd	1.972	2.350	1.476	0.200
s(target_ratio):media3dp	1.047	1.428	1.612	0.268
s(target_ratio):setset1	1.005	1.009	0.650	0.422
s(target_ratio):setset2	1.022	1.042	0.165	0.719
s(user_id)	217.826	258.000	6.423	0.000

Table 5.4: Parametric coefficients in gam model without random guessers.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	3.773	0.083	45.613	0.000
setset2	0.014	0.080	0.172	0.863
media3dd	-0.317	0.077	-4.094	0.000

	Estimate	Std. Error	t value	Pr(>|t|)
media3dp	-0.389	0.089	-4.359	0.000
setset2:media3dd	0.018	0.110	0.163	0.871
setset2:media3dp	0.078	0.124	0.630	0.529

Table 5.5: Approximate significance of smooth terms in gam model without random guessers.

Smooth	edf	Ref.df	F	p-value
s(target_ratio):media2dd	1.002	1.003	0.058	0.813
s(target_ratio):media3dd	1.972	2.350	1.476	0.200
s(target_ratio):media3dp	1.047	1.428	1.612	0.268
s(target_ratio):setset1	1.005	1.009	0.650	0.422
s(target_ratio):setset2	1.022	1.042	0.165	0.719
s(user_id)	217.826	258.000	6.423	0.000

Table 5.6: Estimated marginal means for generalized additive model

Data filtering	set	emmean	SE	df	lower.CL	upper.CL
All participants	set1	3.773	0.083	3729.128	3.611	3.936
		3.532	0.090		3.356	3.709
		3.493	0.105		3.286	3.700
	set2	3.788	0.083		3.625	3.951
		3.565	0.091		3.387	3.742
		3.586	0.106		3.378	3.794
Random guessers removed	set1	3.593	0.109	2814.776		3.807
		3.150	0.107		2.939	3.360
		3.199	0.115		2.973	3.425
	set2	3.693	0.109		3.479	3.906
		3.361	0.108		3.149	3.574
		3.310	0.116		3.081	3.538

Table 5.7: Pairwise differences for generalized additive model

Data filtering	set	contrast	estimate	SE	df	t.ratio	p.value
All participants	set1	2dd - 3dd	0.241	0.088	3729.128	2.736	0.017
		2dd - 3dp	0.280	0.103		2.711	0.018
		3dd - 3dp	0.039	0.109		0.361	0.931
	set2	2dd - 3dd	0.223	0.089		2.516	0.032
		2dd - 3dp	0.202	0.104		1.940	0.128
		3dd - 3dp	-0.021	0.110		-0.192	0.980
Random guessers removed	set1	2dd - 3dd	0.443	0.091	2814.776	4.868	0.000
		2dd - 3dp	0.394	0.100		3.925	
		3dd - 3dp	-0.049	0.099		-0.499	0.872
	set2	2dd - 3dd	0.331	0.092		3.585	0.001
		2dd - 3dp	0.383	0.102		3.757	
		3dd - 3dp	0.052	0.100		0.521	0.861

Chapter 6

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